

Final Report

Adaptive Drift Detection and Response Framework for Financial Machine Learning Models

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Summary

A machine learning model assumes the future will look like the past. In financial markets, that assumption breaks regularly. When it does, a model keeps predicting with the same confidence as before. It just starts being wrong more often, with no way of knowing it.

This project asks a simple question. Can a model tell when the world has changed around it, and adapt fast enough to matter?

The answer, across 4,312 experimental runs on 50 financial instruments spanning nearly a decade, is mostly yes. The detectors correctly identified major market disruptions. Adapting improved both accuracy and risk-adjusted returns, and those gains survived realistic transaction costs. And the simplest possible detector outperformed every more complex version. The project is equally honest about the limits. Retraining improves predictions but damages probability estimates. Large drawdowns become more likely when the framework fires in the wrong direction. And the 2022 rate hiking cycle exposed a hard architectural ceiling that no parameter adjustment could fix.

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Chapter 1

Introduction and Background Research

1.1 Introduction

A predictive model assumes that the future will resemble the past. This is called stationarity. That assumption is reasonable in most supervised learning settings but not in financial markets. Conditions alternate between qualitatively different regimes: bull markets, recessions, inflation cycles, liquidity crises. Each produces a different relationship between observable features and the direction of the next trading day's return. A model calibrated on one regime does not simply lose accuracy in another. It becomes wrong in a way it cannot itself detect. This failure of stationarity is known as *concept drift* [58, 23].

The problem this project addresses is not whether drift exists. The evidence on that is clear. The question is whether drift can be detected early enough to make a difference. Can a model identify that its training distribution has shifted before prediction quality has degraded? And if it can, can it trigger a proportionate response rather than waiting for accuracy to collapse? The framework developed here tests this hypothesis concretely. It monitors the model's rolling prediction loss via a Page Hinkley sequential change point detector and asks whether three supplementary feature distribution detectors (KS, PSI, JS) provide a warning even earlier. A tiered controller triggers one of four retraining responses scaled to severity when a distributional shift is identified.

The empirical findings, reported in Chapter 4, confirm that the early warning hypothesis partially holds. Feature distribution alarms fire approximately 11 days before accuracy deterioration becomes observable, and adaptive retraining based on those alarms produces statistically significant improvements in both directional accuracy and risk adjusted returns. But the findings also confirm the limits of this approach. The Page Hinkley performance monitor alone is the minimum sufficient configuration. Why feature detectors add no lead time is a central analytical result: covariate shift in financial returns rarely precedes the model's loss degradation by more than a few days, and the Page Hinkley monitor's cumulative sum formulation catches persistent shifts almost as quickly. The full feature distribution stack adds no accuracy improvement over Page Hinkley alone. The framework fails during the 2022 Federal Reserve tightening cycle due to an architectural ceiling in batch retraining. The probability calibration degrades as a consequence of the calibration discrimination trade off, a structural feature of the retraining mechanism that is explained in full in Chapter 2.

The rest of this chapter builds the theoretical foundation for those findings. Section 1.2 establishes the empirical evidence that financial markets are non stationary. Section 1.3 gives the formal taxonomy of drift types, covering covariate shift, concept drift, and prior probability shift, and explains why this taxonomy motivates a multi layer detection stack rather than a single signal. Section 1.4 surveys existing detection methods and explains why each of the four selected detectors is included and why the alternatives are rejected. Section 1.5 reviews the adaptive learning literature that constrains the update strategy. Section 1.6 surveys the

financial machine learning evidence that informs model selection and identifies the calibration discrimination trade off as a first class constraint. The three research questions and contributions are stated in Section 1.7.

1.2 Non Stationarity in Financial Time Series

1.2.1 Evidence for Regime Changes

The assumption that the joint distribution $P(\mathbf{x}_t, y_{t+1})$ is stationary is foundational to most supervised learning frameworks. It is empirically untenable in financial time series. Three strands of evidence establish this directly.

The first is structural. Hamilton [28] modelled the business cycle as a two state hidden Markov chain, estimating mean regime durations of 10 to 12 months for expansionary periods and 4 to 6 months for contractionary periods. These are persistent structural shifts, not continuous Gaussian variation. The implication for a predictive model is direct: any training window longer than roughly four months risks spanning two qualitatively different data generating processes.

The second strand is cross asset. Ang and Timmermann [1] showed that equity bond correlations switch sign between regimes. This became directly observable in the 2022 Federal Reserve tightening cycle, where both asset classes sold off simultaneously, breaking a decade long correlation pattern. A static model trained before 2022 would have learnt the wrong joint distribution for the period that followed.

The third strand is volatility driven. Autoregressive Conditional Heteroskedasticity dynamics, known as ARCH and GARCH, describe how large price moves cluster in time: high volatility tends to be followed by more high volatility, and calm periods by more calm [16, 7]. These models imply that high volatility periods cluster across asset classes, so a model trained during a calm regime will encounter a categorically different input distribution during a volatility spike. The rolling KS statistic for four representative instruments (AAPL, SPY, GLD, BTC USD) over 2016 to 2024 in Figure B.10 (Appendix B) confirms that sustained elevations coincide precisely with the COVID 19 dislocation, the 2022 Federal Reserve hiking cycle, the Silicon Valley Bank collapse, and a 2021 cryptocurrency regime shift.

Taken together, these three strands establish that distributional shift in financial time series is not a rare edge case. It is a recurring, temporally clustered phenomenon that any deployed model will encounter.

1.2.2 The Adaptive Markets Hypothesis

The standard theoretical response to this evidence is the Efficient Markets Hypothesis (EMH) [18]. In its weak form, the EMH asserts that all historical price information is already incorporated into current prices. If the EMH held strictly, no model trained on historical features could predict future returns, and the premise of this project would be false right from the beginning.

Lo's Adaptive Markets Hypothesis (AMH) [40] provides a more empirically consistent account. The AMH treats market participants as adaptive agents who learn from experience, so the

degree of market efficiency varies over time and across regimes. Predictable patterns emerge when conditions change faster than participants can adapt, and disappear once adaptation catches up. The challenge is therefore not whether predictability exists, but detecting when the distribution has shifted and adapting before the market reaches its new equilibrium. The AMH is what makes the early warning problem tractable: if detection is fast enough, the adaptive model can exploit the post shift window before equilibrium is restored.

1.3 Concept Drift: Taxonomy and Formal Definitions

Knowing that financial distributions shift is a necessary motivation for this project, but not a wholly sufficient one. It is necessary to be precise about what shifts to design a detection stack. Different types of distributional change require different detection signals, and no single detector covers them all.

1.3.1 Formal Definitions

Let $P_t(\mathbf{x}, y)$ denote the joint distribution of feature vector $\mathbf{x}_t \in \mathbb{R}^d$ and binary target $y_t \in \{0, 1\}$ at time t . Concept drift in its general form is any change in this joint distribution:

$$\exists t_1 < t_2 : P_{t_1}(\mathbf{x}, y) \neq P_{t_2}(\mathbf{x}, y) \quad (1.1)$$

Gama et al. [23] decompose this into three distinct types based on which marginal or conditional distribution has shifted. The decomposition matters because it determines what signal a detector needs to monitor.

Covariate shift occurs when the marginal distribution of input features changes, but the predictive relationship between features and the target remains approximately stable:

$$P_{t_1}(\mathbf{x}) \neq P_{t_2}(\mathbf{x}), \quad P_{t_1}(y | \mathbf{x}) \approx P_{t_2}(y | \mathbf{x}) \quad (1.2)$$

In financial data, this is what happens at the onset of a volatility regime change. Feature distributions shift substantially before the predictive relationship has changed. A model experiencing covariate shift receives out of distribution inputs but is not yet systematically wrong. Error based detectors will not catch it at this stage. Distribution based detectors can.

Concept drift (in the strict sense) occurs when the predictive relationship itself changes:

$$P_{t_1}(y | \mathbf{x}) \neq P_{t_2}(y | \mathbf{x}) \quad (1.3)$$

This is the most damaging form because it directly invalidates the learnt mapping. In financial markets it accompanies structural breaks such as a central bank policy reversal, where the same features that predicted an up day now predict a down day.

Prior probability shift occurs when the underlying proportion of up days versus down days changes, while the features that distinguish the two classes remain structurally similar:

$$P_{t_1}(y) \neq P_{t_2}(y), \quad P_{t_1}(\mathbf{x} \mid y) \approx P_{t_2}(\mathbf{x} \mid y) \quad (1.4)$$

In practice the three types co occur. The 2022 Federal Reserve tightening produced simultaneous shifts in feature distributions, the feature label relationship, and the proportion of up days. This exposes the taxonomy’s tripartite separation as a modelling convenience rather than an empirical reality. What matters for design is the key implication established by Widmer and Kubat [58]: no single error based detector can resolve which component has shifted, because covariate shift is invisible to any detector that only watches the error stream. This is the direct motivation for a multi layer stack rather than a single detector. Yet as Chapter 4 shows, the additional detectors prove redundant in practice: the Page Hinkley monitor’s sensitivity to persistent loss increases catches the shift almost as early as any feature distribution alarm, because financial covariate shifts are rarely pure and almost always accompanied by small but cumulative prediction errors.

1.4 Drift Detection Methods

1.4.1 Categories of Drift Detector

Drift detectors partition into three categories based on what they monitor. The choice is not primarily a performance question. It is a question of what information is available at detection time.

Error based detectors monitor the model’s prediction error stream. They carry two critical limitations for this setting. First, the true label y_{t+1} must be available to compute per step error, making them inapplicable when labels arrive with a one day lag. Second, they are blind to covariate shift that has not yet propagated to the error stream. Given that covariate shift precedes error degradation by a mean of 11.1 days (Chapter 4), this is not a minor limitation.

Distribution based detectors monitor the statistical properties of the input feature distribution without requiring labels, enabling detection before accuracy degrades. Their limitation is that a shift in $P(\mathbf{x})$ does not guarantee a change in $P(y \mid \mathbf{x})$. This is addressed by asset adaptive calibration, described in Chapter 2, Section 2.4.4.

Model based detectors monitor output behaviour rather than inputs. The prediction drift monitor applied here uses JS divergence on the output probability distribution. This covers cases where the model’s confidence profile changes without a corresponding shift in the marginal feature distributions.

1.4.2 Principal Detection Methods: Survey and Critique

Rabanser et al. [51] systematically compare ten methods on tabular data, finding that PSI class statistics achieve near identical detection power to KS at $\mathcal{O}(n)$ versus $\mathcal{O}(n \log n)$ cost, and that the KS test retains value only for tail sensitivity. The four selected methods (KS, PSI, JS, and

Page Hinkley) are described in detail in Chapter 2, Section 2.3. Their properties are summarised in Table 1.1 below.

ADWIN. ADWIN [6] uses a Hoeffding bound test on a variable length window but requires the true label, making it inapplicable under a one day lag. It is also blind to covariate shift in $P(\mathbf{x})$. The synthetic benchmark in Chapter 3 confirms the consequence: ADWIN achieves 0.2% TPR on feature drift tasks where PSI+PH achieves 93 to 100%.

DDM, EDDM, and Kifer et al. DDM [22] and EDDM [3] share ADWIN’s label dependency and covariate shift blindness [46]. Kifer et al. [31] generalise KS and PSI under a common divergence framework, but Rabanser et al. [51] show that PSI class methods already achieve near optimal sensitivity on OHLCV type tabular drift, leaving no motivation for the added configuration overhead here.

1.4.3 Comparative Analysis

Table 1.1 summarises the principal methods along the four dimensions most relevant to the deployment constraints established in Chapter 2.

Table 1.1: Comparative analysis of drift detection methods. "Requires labels" indicates whether the true target y_{t+1} must be available at detection time. "Bounded" indicates whether the statistic has a natural $[0, 1]$ or otherwise finite range facilitating aggregation. PH is the primary (minimum sufficient) detector; KS, PSI, and JS are supplementary probes testing the lead time hypothesis (Chapter 2, Section 2.3).

Method	Type	Requires labels	Bounded	Complexity	Role
Page Hinkley	Performance	No	No (unbounded)	$\mathcal{O}(1)$	Primary
KS test	Distribution	No	Yes $[0, 1]$	$\mathcal{O}(n \log n)$	Supplementary
PSI	Distribution	No	No (but practical thresholds exist)	$\mathcal{O}(n)$	Supplementary
JS divergence	Distribution	No	Yes $[0, \ln 2]$	$\mathcal{O}(n)$	Supplementary
ADWIN	Error based	Yes	No	$\mathcal{O}(\log n)$	Rejected
DDM / EDDM	Error based	Yes	No	$\mathcal{O}(1)$	Rejected
Kifer et al.	Distribution	No	Depends on divergence used	$\mathcal{O}(n)$	Rejected

The key asymmetry in the table is between the first four rows and the last three. The selected detectors require no labels and cover covariate shift, concept drift, and prior probability shift between them. The rejected detectors either need labels (ADWIN, DDM, EDDM), failing requirement 1, or add overhead over PSI class methods with no sensitivity gain (Kifer et al.). The acknowledged limitations of the selected stack are acceptable given that financial regimes span weeks to months [28]. PSI is bin count sensitive. KS is $\mathcal{O}(n \log n)$. PH detects only persistent shifts [47]. These are quantified directly in the synthetic benchmark of Chapter 3.

1.5 Adaptive Learning Under Drift

Detecting drift is only half the problem. Once an alarm fires, the model must update in a way that is responsive to the shift without discarding everything it has learnt. The adaptation literature constrains this choice in three ways.

Klinkenberg and Joachims [33] show that sliding window retraining is effective but that the optimal window size varies with drift type. The right window for an abrupt shock is not the right window for a gradual tightening cycle. This motivates asset adaptive calibration of the window and threshold parameters rather than a fixed global setting.

Herbster and Warmuth [29] establish that a blended learner achieves regret $\mathcal{O}(\sqrt{T \ln N})$, where T is the number of time steps and N is the number of experts in the blend. This bound suggests retaining prior model parameters rather than replacing them fully at each update, a soft update that sits closer to the stable end of the stability plasticity spectrum. The specific blend weight $\gamma = 0.15$ is justified in Chapter 2, Section 2.4.6. Losing et al. [41] provide empirical confirmation that blended approaches outperform both static and full retrain strategies under gradual drift.

Dynamic ensemble methods such as DWM [35], ARF [25], and DDD [43] adapt continuously through ensemble diversity rather than discrete retrain triggers, and represent a credible alternative in principle. They are excluded here on two independent grounds. First, using dynamic ensembles in both the adaptive and static arms would confound the detection effect, making it impossible to attribute any measured improvement to the detection mechanism rather than the ensemble architecture. Second, ARF and DDD require a continuous observation stream that is inconsistent with the 400 row minimum training window required for reliable calibration on sparse regime change events.

1.6 Machine Learning in Financial Prediction

Two modelling decisions are open at this point. Firstly, which base classifier to use, and whether the task is feasible at all. Both are addressed by the financial machine learning literature.

Gu et al. [27] and Krauss et al. [36] establish Random Forest and Gradient Boosted Machine as the best performing classifiers for next day equity return direction, which is the same task used here. Volatility and momentum features are most persistently informative across market conditions. Both studies also note that predictive performance is itself non stationary. The Fama French factor literature [19] shows that momentum and volatility premia flip sign during structural breaks, which is precisely why a richer static feature set alone cannot solve the problem. Rapach and Zhou [52] frame the core challenge concisely: signals exist but are time varying, so the task is detecting when they have stopped working.

One further implication follows from DeGroot and Fienberg [12] and constrains the entire adaptation strategy. Retraining a classifier on new data shifts its decision boundary in a way that can improve directional accuracy while simultaneously degrading probability calibration, because the boundary moves without a corresponding recalibration step. This calibration discrimination trade off is not a bug in the implementation. It is a structural consequence of

the adaptation mechanism, and it is documented as a limitation in Chapter 4, Section 4.5.

1.7 Research Questions and Contributions

1.7.1 Research Questions

The literature reviewed above converges on a specific gap. Non stationary financial markets (Section 1.2) demand detection before accuracy degrades. The drift taxonomy (Section 1.3) shows that covariate shift is invisible to error based detectors. The adaptation literature (Section 1.5) constrains the update to a blended strategy with asset adaptive parameters. The financial machine learning evidence (Section 1.6) confirms the base classifier choices and flags the calibration discrimination trade off as unavoidable. No existing study integrates all four constraints into a single validated framework. Three research questions operationalise this gap:

1. **RQ1: Do distribution based detectors identify genuine market regime changes?** Does the detector stack produce alarms that correspond to dated market events in the literature, and is adaptive accuracy improvement concentrated in alarm active periods?
2. **RQ2: Does adaptive retraining improve prediction quality?** Does the improvement hold across 50 instruments and four market sub periods, and is it statistically robust and economically meaningful after transaction costs?
3. **RQ3: What is the minimum sufficient detector configuration?** Is the full four detector stack necessary, or does a smaller subset achieve equivalent performance at lower computational cost?

1.7.2 Contributions

This project makes three contributions.

The primary contribution is a validated adaptive drift detection framework for financial binary classification that combines label free detection, asset adaptive calibration, and a tiered retraining controller. Prior detection focused work [51, 6] evaluates detectors in isolation. Prior financial machine learning work [27, 36] acknowledges non stationarity without providing a systematic detection and adaptation mechanism. The synthetic benchmark in Chapter 3 and the 50 instrument, ten year evaluation in Chapter 4 address both gaps.

The second contribution is the composite drift index: a Page Hinkley performance monitor as the minimum sufficient detector, supplemented by three feature distribution signals (KS, PSI, JS) that test whether upstream monitoring provides earlier warning. The ablation result shows the hypothesis is rejected. PH alone is Pareto optimal. But the index with asset adaptive calibration generalises to any setting where fixed global thresholds are inappropriate.

The third contribution is a fully reproducible implementation: 121 tests, 23 experiment scripts with committed results, and a real time monitoring dashboard, all version controlled and publicly available.

Chapter 2 now takes these theoretical foundations and turns them into concrete, justified design decisions.

Chapter 2

Methodology

Chapter 1 established the problem and the theoretical tools to approach it. Financial markets shift between qualitatively different regimes. No single detector covers all three types of distributional change. Blended updates outperform both static models and full retraining. This chapter turns those conclusions into a working system. Each design choice here connects to a specific requirement, theoretical result, or empirical finding from Chapter 1.

2.1 Problem Formulation

2.1.1 Binary Classification Task

The task is predicting whether a financial instrument will close higher or lower on the next trading day. Let C_t denote the closing price on trading day t , and $\mathbf{x}_t \in \mathbb{R}^d$ the feature vector constructed from market data available at the close of day t . The next day log return, which is additive over time so that multi day returns are simply their sum, is:

$$r_{t+1} = \log \frac{C_{t+1}}{C_t} \quad (2.1)$$

The binary classification target is the sign of this return:

$$y_{t+1} = \mathbf{1}[r_{t+1} > 0] \in \{0, 1\} \quad (2.2)$$

A classifier $f_\theta : \mathbb{R}^d \rightarrow [0, 1]$ estimates the probability of an up day: $\hat{p}_{t+1} = P(y_{t+1} = 1 \mid \mathbf{x}_t)$.

The binary label follows by thresholding at 0.5.

Binary classification is chosen over regression for three reasons, each specific to the financial setting. First, directional accuracy determines whether to go long or short on a given day. A model that predicts $r_{t+1} = +0.3\%$ when the true return is $+1.2\%$ still makes the correct trading decision, whereas mean squared error treats this as a large penalty [27]. Second, financial return distributions have extremely heavy tails. A single crash day return can be 10 to 20 standard deviations from the mean. Under a regression objective, these outliers dominate the loss and destabilise training, whereas a binary target is bounded by construction and unaffected by the magnitude of the return [52]. Third, the Information Coefficient framework used for evaluation in Section 4.3.5 is defined over rank correlations between predicted probabilities and realised directions, making a probabilistic binary output a natural fit [26].

2.1.2 The Non Stationarity Problem

A static classifier assumes the joint distribution $P(\mathbf{x}_t, y_{t+1})$ is fixed. In financial markets it is not. Structural breaks, volatility regime transitions, and microstructure changes all shift this distribution in ways that invalidate a model trained on earlier data [28, 40]. All three drift types from Chapter 1 arise here. A framework that cannot respond will accumulate prediction error silently.

2.1.3 Sequential Evaluation Protocol

Look ahead bias is the central pitfall in financial backtesting. It means using future information to make predictions about the past. It is bad because the resulting accuracy figures look plausibly good rather than obviously wrong. A model that inadvertently uses tomorrow's opening price to predict tomorrow's direction will achieve near perfect accuracy in simulation but will be useless in live trading. The academic and practitioner literatures are full of backtested strategies that appeared exceptional in simulation and failed completely in deployment because of exactly this error [52]. Four protocol rules prevent it here.

First, every feature on day t uses only data available at the close of day t . Second, the train/test split is strictly sequential: the model trains on the three calendar years before each sub period start (approximately 756 trading days), across four evaluation windows: *early* (2015 to 2019), *std* (2018 to 2024), *covid* (2020 to 2024), and *recent* (2022 to 2024) [28, 1]. Third, the static baseline's parameters are frozen at the end of training and never updated. Fourth, every adaptive retrain uses only data available at the time of the trigger. No future data is consulted to determine whether the retrain was beneficial. Compliance with all four rules is verified programmatically by the data layer unit test described in Chapter 3, Section 3.6.1.

2.2 Data Collection and Preparation

2.2.1 Data Sources

Price and volume data for all 50 instruments were sourced from Yahoo Finance via `yfinance`. Daily OHLCV data (Open, High, Low, Close, Volume) were downloaded from January 2010 to December 2024. Macro series features were sourced from FRED via `fredapi`. Both sources are publicly available and used under their respective terms of service (Appendix B).

2.2.2 Feature Engineering

All features use only OHLCV data available before the close of day t . The vector $\mathbf{x}_t \in \mathbb{R}^d$ comprises six groups.

Return features. Simple return $\rho_t = (C_t - C_{t-1})/C_{t-1}$ and log return $r_t = \log(C_t/C_{t-1})$.

Volatility features. Rolling realised volatility over four lookback windows:

$$\sigma_{\tau,t} = \sqrt{\frac{1}{\tau-1} \sum_{i=0}^{\tau-1} (r_{t-i} - \bar{r}_{\tau,t})^2}, \quad \tau \in \{5, 10, 20, 60\} \quad (2.3)$$

The four horizons capture intra week noise (5 days), mean reversion dynamics (10 days), the GARCH volatility half life (where GARCH, Generalised Autoregressive Conditional Heteroskedasticity, describes how financial volatility clusters so that calm periods tend to follow calm periods and turbulent ones tend to follow turbulent ones [16, 7]) (20 days), and Hamilton regime onset [28] (60 days), consistent with Gu et al. [27].

Momentum features. Cumulative log return over the same four windows:

$m_{\tau,t} = \sum_{i=0}^{\tau-1} r_{t-i}$, $\tau \in \{5, 10, 20, 60\}$. A positive $m_{\tau,t}$ means prices have been rising over τ days.

Drawdown. Rolling maximum drawdown captures sustained price declines independently of volatility:

$$DD_{\tau,t} = \frac{C_t}{\max_{i \in [t-\tau+1, t]} C_i} - 1, \quad \tau \in \{5, 10, 20, 60\} \quad (2.4)$$

$DD_{\tau,t} \leq 0$ always. A value of zero means today's close equals the τ day high.

OHLCV ratio features. Three scale invariant features summarise intra day price and volume structure:

$$HL_t = \frac{H_t - L_t}{C_t}, \quad CO_t = \frac{C_t - O_t}{O_t}, \quad VChg_t = \frac{V_t - V_{t-1}}{V_{t-1}} \quad (2.5)$$

Volume Z score and trend regime features. The volume Z score measures how unusual today's trading activity is relative to recent history. High volume during a price move tends to signal institutional participation and sustained regime transitions [10]:

$$Z_{\tau,t}^V = \frac{V_t - \bar{V}_{\tau,t}}{\sigma_{\tau,t}^V}, \quad \tau \in \{5, 10, 20, 60\} \quad (2.6)$$

Two trend features measure where today's close sits relative to its rolling mean in standard deviation units. This is a price Z score that identifies whether the instrument is in an extended trend:

$$Z_{\tau,t}^P = \frac{C_t - MA_{\tau,t}}{\sigma_{\tau,t}}, \quad \tau \in \{20, 60\}; \quad MACross_t = \frac{MA_{20,t} - MA_{60,t}}{MA_{60,t}} \quad (2.7)$$

A positive $MACross_t$, meaning the 20 day mean sits above the 60 day mean, is the classic golden cross upward alignment [10].

The full feature vector has dimension $d=33$. The look ahead prevention test in Section 3.6.1 verifies no feature uses data from day $t+1$ or later.

2.2.3 Preprocessing and Quality Checks

Features are scaled using **RobustScaler**, which centres on the median and scales by the interquartile range (IQR) [30, 54]:

$$\tilde{x}_{j,t} = \frac{x_{j,t} - \text{median}(x_j)}{\text{IQR}(x_j)} \quad (2.8)$$

Financial returns are leptokurtic, which means their tails are heavier than a Gaussian. A single crash day return can inflate the standard deviation and compress all other values toward zero.

The median and IQR are resistant to such outliers by construction, making **RobustScaler** more appropriate than **StandardScaler** here [16, 7]. Median and IQR are computed on the training window only. Missing prices are forward filled. Non finite warm up values are set to

zero. Instruments with more than 5% missing values or fewer than 1,200 clean trading days were excluded. None of the 50 instruments triggered either threshold.

2.3 Detector Design Rationale

2.3.1 Requirements Driving Detector Selection

Choosing a detector depends on what information is available at detection time and what type of shift must be caught. Four requirements follow from the problem structure.

1. **Label free operation.** The true label y_{t+1} is unavailable until the next trading day. Any detector requiring labelled feedback cannot monitor drift before accuracy degrades.
2. **Coverage of all three drift types.** A single detector misses shift types that do not propagate to its signal [23]. A composite stack is required.
3. **Normalisable output.** All component scores must be normalisable to $[0, 1]$ for aggregation into the composite drift index.
4. **Computational feasibility.** At least one detector must operate in $\mathcal{O}(1)$ time and space to keep streaming extensions feasible.

2.3.2 Selected Detectors and Their Roles

Page Hinkley (PH) [47, 44] is the minimum detector satisfying all four requirements. It detects persistent upward shifts in rolling log loss (a penalty score that increases the more confidently the model is wrong) via a CUSUM formulation (Cumulative Sum: a technique that accumulates small deviations over time so that a sustained shift eventually crosses a threshold, rather than reacting to any single spike), operates in $\mathcal{O}(1)$ time and space, and requires only predicted probabilities. Whether monitoring upstream feature distributions provides earlier warning is an empirical question. Three supplementary detectors test this. Results are in Chapter 4, Section 4.4.1. As Chapter 4 shows, PH alone suffices because the cumulative sum of log loss deviations catches the shift almost as early as any feature distribution change. The additional lead time from KS, PSI, or JS is negligible in financial data, where persistent prediction errors begin accumulating within days of a regime shift.

Kolmogorov Smirnov (KS) [34, 57] measures the largest gap between two empirical cumulative distribution functions, capturing how differently two samples are distributed. It is bounded in $[0, 1]$, non parametric, and addresses covariate shift in $P(\mathbf{x})$.

Population Stability Index (PSI) [56, 4] compares the proportion of observations in each decile bin between the reference and current windows. Developed in credit risk monitoring, it has established threshold conventions: PSI below 0.10 is stable, 0.10 to 0.25 is moderate shift, and 0.25 or above is significant. It captures smooth population shifts that KS's supremum statistic may miss. Decile binning (10 bins) is standard in Basel II monitoring and keeps expected bin counts above five in a 100 row window.

Jensen Shannon divergence (JS) [39, 55] is applied to the model's output probability distributions, monitoring prediction drift. JS is preferred over KL divergence because it is symmetric and bounded in $[0, \ln 2]$, enabling normalisation to $[0, 1]$.

ADWIN [6], DDM [22], and EDDM [3] require the true label at detection time, failing requirement 1. They are also blind to covariate shift, failing requirement 2. The synthetic benchmark confirms the consequence: ADWIN achieves 0.2% true positive rate on feature drift tasks where PSI+PH achieves 93 to 100%. Kifer et al. [31] add overhead over PSI class methods with no sensitivity gain [51]. The full comparison is in Table 1.1 (Chapter 1).

2.4 Parameter Justification

2.4.1 Rolling Window Size: `window_size = 100`

At $n=100$ and $\alpha=0.05$, the KS critical value $D_{0.05} \approx 0.192$ detects Gaussian mean shifts of approximately 0.5σ , which is the typical magnitude at regime onset. Reducing to $n=60$ raises the critical value to 0.248, missing sub 0.7σ shifts. Hamilton [28] estimates short US regimes at 60 to 80 trading days, so a 100 day window stays within a single regime while satisfying the statistical power bound. Extending beyond 150 days risks spanning two consecutive regimes, contaminating the reference distribution with pre and post shift observations and reducing detection sensitivity. Sensitivity analysis confirms that the coefficient of variation (CV, defined as standard deviation divided by mean, measuring relative variability across the grid) of accuracy delta is below 3 for 13 of 16 tickers across the 100 to 150 day range (Chapter 4, Section 4.6).

2.4.2 Cooldown Period: `cooldown_days = 5`

After any retrain, alarm signals are suppressed for 5 trading days. MiFID II Article 5 [17] mandates T+2 settlement for EU equity markets, so a 5 day cooldown provides a 2.5 cycle buffer before conflicting position changes can arise. Separately, 43% of low accuracy days fall within 5 days of a prior drift event (Figure B.20, Appendix B), so the window also prevents redundant retrains during post drift turbulence.

2.4.3 Composite Drift Index Weights: `0.4 / 0.3 / 0.3`

$$\mathcal{D}_t = 0.4 \cdot s_{\text{feature},t} + 0.3 \cdot s_{\text{prediction},t} + 0.3 \cdot s_{\text{performance},t} \quad (2.9)$$

The weights reflect causal ordering. A shift in $P(\mathbf{x})$ is upstream of a shift in $P(\hat{y} \mid \mathbf{x})$, which is upstream of $P(\text{loss})$ degradation [23]. The earliest signal receives the greatest weight. Weights were set heuristically before any experiments ran.

2.4.4 Calibration Percentiles: `60th / 75th / 90th / 95th`

The four action tiers map to the 60th, 75th, 90th, and 95th percentiles of each asset's reference window drift index. A fixed global threshold is inappropriate: a 310 basis point (one basis point is 0.01% of value, so 310 basis points is a 3.1% move) daily move in Bitcoin is routine, whereas the same move in a US Treasury ETF signals a rate shock. Data driven, asset specific thresholds outperform fixed ones under non stationarity [6]. When the reference window contains fewer than 50 observations, the module falls back to default limits (low = 0.6, moderate = 1.0, high = 1.5, severe = 2.0).

2.4.5 Exponential Moving Average (EMA) Smoothing: $\alpha = 0.10$

$$\tilde{\mathcal{D}}_t = 0.10 \cdot \mathcal{D}_t + 0.90 \cdot \tilde{\mathcal{D}}_{t-1} \quad (2.10)$$

An EMA applies more weight to recent observations and less to older ones, controlled by the smoothing parameter α . At $\alpha = 0.10$ the EMA half life is approximately 6.6 trading days, suppressing single day artefacts from fat tailed returns without masking genuine sustained shifts [24, 11]. This timescale is mutually consistent with the 5 day cooldown: a post drift turbulence spike dissipates before the next alarm window opens.

2.4.6 Weighted Ensemble Blend: $\gamma = 0.15$

$$\theta_{\text{new}} = (1 - \gamma) \theta_{\text{old}} + \gamma \theta_{\text{recent}} \quad (2.11)$$

Setting $\gamma = 0.15$ blends 85% of prior parameters with 15% of recent window parameters. This convex combination is well defined for linear models whose parameters are real-valued vectors; for tree-based ensemble models (Gradient Boosting, Random Forest), direct parameter blending is not applicable, and adaptation takes the form of a full batch retrain on the rolling window. In the SGD strategy, γ governs a prediction-level blend between the online logistic regression component and the primary tree-based predictor. The Herbster and Warmuth [29] regret bound $\mathcal{O}(\sqrt{T \ln N})$ motivates retaining prior parameters rather than full replacement at each update. The value is consistent with Hamilton’s [28] finding that financial regimes persist for months, so historical data retains predictive value after a regime transition. In the retrain ablation (Chapter 4, Table 4.2), the SGD weighted update at $\gamma=0.15$ achieves the highest ΔSharpe of +0.128. The Sharpe ratio measures annualised return divided by annualised return volatility; ΔSharpe is the adaptive model’s Sharpe minus the static model’s Sharpe, so a positive value means the adaptive model earns more return per unit of risk.

2.4.7 Position Gate Weights: 0.60 / 0.40

Rather than switching between static and adaptive models with a hard binary switch, which would amplify calibration errors whenever the adaptive model is activated in a non drift period, the position gate blends their outputs continuously as a function of detected drift severity:

$$w_{\text{adapt}} = \text{clip}\left(0.60 \cdot \tilde{\mathcal{D}}_t + 0.40 \cdot \tilde{s}_{\text{perf},t}, 0.25, 0.90\right) \quad (2.12)$$

The 0.60/0.40 split reflects causal asymmetry. The drift signal $\tilde{\mathcal{D}}_t$ fires at lag zero relative to the distributional change, monitoring upstream feature distributions that shift before accuracy degrades. The performance signal $\tilde{s}_{\text{perf},t}$ lags by 11.0 days on average, rising only after the error stream has accumulated evidence of degradation. Weighting $\tilde{\mathcal{D}}_t$ at 0.60 and $\tilde{s}_{\text{perf},t}$ at 0.40 is therefore the natural causal default, supported post hoc by the Chapter 4 results: drift active days produce a $4.6\times$ larger mean accuracy delta than quiet days (BCa 95% CI [+0.0082, +0.0116], Section 4.2.3), where BCa denotes bias-corrected and accelerated bootstrapping, correcting for skew in the underlying distribution.

The clip bounds [0.25, 0.90] prevent degenerate outcomes: the adaptive model always contributes at least 25% and the static model retains at least 10%, so neither is fully discarded. The static model’s parameters remain frozen throughout; only the blend weight applied to its

output varies with drift severity. The split was not grid searched, as data-driven weight optimisation is unavailable within the strictly sequential evaluation protocol and is noted as a limitation in Section 4.5.

2.4.8 Minimum Retrain Rows and Out-of-Sample (OOS) Rollback

At 252 trading days per year, `min_retrain_rows` = 400 corresponds to approximately 19 months. This is the minimum needed to avoid forcing pre shift data into every retrain. It is an architectural constraint, not a tunable parameter. Section 4.6 confirms values from 200 to 500 rows produce identical 2022 degradation. After each retrain, the new model is validated on the 20 most recent rows that were not used for training (the out-of-sample holdout). If accuracy falls more than 8 percentage points below the pre retrain model, the controller reverts to the previous snapshot.

2.5 Model Selection

Three base classifiers span the bias variance frontier. Random Forest (RF) reduces variance through bagging, which means training many decision trees on random subsets of data and averaging their predictions so that no single noisy tree dominates the output. Gradient Boosted Machine (GBM) reduces bias through residual fitting, meaning each new tree corrects the errors of the previous one and gradually improves accuracy on the training distribution. Logistic Regression (LR) provides an interpretable linear baseline against which the non linear gains of the tree based models can be cleanly measured [8, 21].

Krauss et al. [36] validate RF and GBM as the best performing classifiers for next day S&P 500 return direction, the same task used here. Including all three tests whether the adaptive benefit generalises across inductive biases rather than being specific to one model family. LSTMs were excluded on two independent grounds. First, tree based models outperform them on tabular OHLCV data [36, 52]. Second, mini batch training is structurally incompatible with discrete sliding window retrains, which replace the full training set at each trigger event.

2.6 Project Risk and Version Control

Five principal risks were identified at project inception. Mitigations are embedded in the framework design. Table B.1 in Appendix B summarises each risk, its assessed likelihood and impact, and the corresponding mitigation.

The codebase is managed on GitHub using Git throughout the full development lifecycle, with each component developed on a separate branch and merged into main only after unit tests passed. Commits were made at each meaningful milestone: component completion, experiment runs, and result-changing bug fixes. Every results CSV and figure is committed alongside the code that generated it, meaning any number in Chapter 4 can be traced to the exact script and parameters that produced it, and the full project reproduces from a clean clone with a single command. A Gantt chart of the project timeline is provided in Figure B.28 in Appendix B.

Chapter 3

Implementation and Validation

Chapter 2 discussed the design choices. Each detector was selected for a reason, every parameter was justified from our principles, and project risks were anticipated. Naturally the next question is whether the system built from those choices actually works. This chapter answers that through a overview of the five components, detailed detector implementations, a description of the controller, and a layered validation programme that establishes correctness before any real data experiments are run.

3.1 System Overview

The framework comprises five loosely coupled components, each with a single responsibility. Loose coupling means each component can be tested, modified, and replaced independently. The full pipeline is shown in Figure B.1 (Appendix B).

1. **Data layer** (`src/data_loader.py`): downloads OHLCV and FRED macro series, constructs the 33 feature matrix, and encodes the binary target.
2. **Model layer** (`src/model/`): maintains static and adaptive instances under a common interface for RF, GBM, and LR, each wrapped in a `RobustScaler` scikit learn `Pipeline`.
3. **Detector stack** (`src/drift_detectors/`): five detector classes exposing a common `detect(reference, current)` interface. PH is the primary performance monitor. KS, PSI, JS, and the prediction drift monitor are supplementary probes testing the lead time hypothesis from Chapter 2, Section 2.3.
4. **Controller** (`src/controller/`): aggregates scores into a composite drift index, calibrates asset adaptive thresholds, and dispatches one of four retraining actions.
5. **Evaluation layer** (`src/evaluation/`): records per run outcomes and contains the 23 experiment scripts used in Chapter 4.

Detectors produce raw numerical scores. The controller handles thresholding, weighting, and dispatch, keeping each component independently testable.

3.2 Drift Detector Implementations

Each detector is implemented as an original Python class from the mathematical definitions in the cited papers. No detection logic was copied from external libraries. The PSI implementation reproduces the worked example in Siddiqi [56] to four decimal places as a correctness check.

3.2.1 Kolmogorov Smirnov Test

The KS statistic measures the largest vertical gap between two empirical cumulative distribution functions (CDFs). An empirical CDF is the step function that rises by $1/n$ at each observed data point. The gap between two CDFs at any given point tells you how differently the two samples are distributed up to that value.

To make this concrete: imagine plotting the sorted feature values from a reference window and from the current window as two staircase curves. The KS statistic is the maximum height difference between those two staircases anywhere along the x axis. A large gap means the current distribution has shifted substantially from the reference:

$$D_{n,m} = \sup_x |F_n(x) - G_m(x)| \quad (3.1)$$

where $F_n(x)$ is the reference window CDF and $G_m(x)$ is the current window CDF.

$D_{n,m} \in [0, 1]$, non parametric [34, 57, 16]. The implementation uses a merge walk over two sorted arrays in $\mathcal{O}(n \log n)$. p values are from `scipy.stats.ks_2samp`. KS is applied per feature, and the maximum across all features is the aggregate. The ablation in Chapter 4 (Table B.1, Appendix B) confirms KS is redundant when PSI is active [51].

3.2.2 Population Stability Index

PSI measures how much a population has shifted relative to a reference by comparing the proportion of observations falling in each quantile bin. The intuition is straightforward: if 20% of reference data fell in a certain range and now only 5% does, the population has moved away from that range. PSI captures this movement across all bins simultaneously and sums it into a single score. Because the bins are defined by the reference distribution’s own deciles, each bin holds roughly equal mass in the reference window, making PSI sensitive to shifts anywhere in the distribution and not just in the tails:

$$\text{PSI} = \sum_{b=1}^B (p_b^{\text{current}} - p_b^{\text{ref}}) \ln \frac{p_b^{\text{current}}}{p_b^{\text{ref}}} \quad (3.2)$$

where $B = 10$ decile bins are defined by the reference window quantiles. p_b^{ref} and p_b^{current} are the corresponding bin fractions. Laplace smoothing ($\epsilon = 10^{-6}$) prevents division by zero. PSI is applied per feature. The maximum is the aggregate.

3.2.3 Jensen Shannon Divergence

JS divergence is applied to the model’s output probability distributions, monitoring prediction drift. To understand why JS rather than the more familiar KL divergence: KL divergence measures how much one distribution differs from another, but it is asymmetric ($D_{\text{KL}}(P\|Q) \neq D_{\text{KL}}(Q\|P)$) and can be infinite if one distribution assigns zero probability where the other does not. JS solves both problems by comparing each distribution not directly to the other, but to their mixture. The result is always finite, always symmetric, and bounded in $[0, \ln 2]$, making it safe to normalise for the composite index:

$$\text{JS}(P\|Q) = \frac{1}{2}D_{\text{KL}}\left(P\left\|\frac{P+Q}{2}\right.\right) + \frac{1}{2}D_{\text{KL}}\left(Q\left\|\frac{P+Q}{2}\right.\right) \quad (3.3)$$

where P is the reference window output distribution, Q is the current window output distribution, and $D_{\text{KL}}(A\|B) = \sum_i A_i \ln(A_i/B_i)$ is the Kullback Leibler divergence [39, 55, 38]. Both distributions are discretised into 20 equal width bins. Laplace smoothing prevents log of zero at bin boundaries.

3.2.4 Page Hinkley Sequential Change Point Detector

PH [47, 44] detects a persistent upward shift in the mean of a stream. Applied here, it monitors rolling log loss. Log loss penalises confident wrong predictions heavily: if the model assigns a 95% probability to an up day and the day turns out to be a down day, the penalty is severe. A sustained rise in log loss therefore signals that the model’s calibration is degrading and that its confident predictions are increasingly pointing the wrong way.

PH works by accumulating the deviation of each new observation from the running mean. Rather than reacting to any single large value, which could just be a fat tailed return day, it requires the deviations to compound over time before raising an alarm. This persistence requirement is what makes it reliable in financial data, where individual extreme days are common but sustained deterioration is the signal worth catching. State updates at each observation x_t :

$$\bar{x}_t = \frac{1}{t} \sum_{i=1}^t x_i \quad (3.4)$$

$$U_t = U_{t-1} + (x_t - \bar{x}_t - \delta) \quad (3.5)$$

$$m_t = \min(U_1, \dots, U_t) \quad (3.6)$$

$$\text{PH}_t = U_t - m_t \quad (3.7)$$

where $\delta > 0$ is an asset adaptive sensitivity parameter that prevents small fluctuations from accumulating in U_t . The alarm fires when PH_t exceeds the per asset threshold λ . Each update requires $\mathcal{O}(1)$ time and space, maintaining five scalar state variables. PH is the only detector suitable for genuine streaming at tick resolution.

3.2.5 Prediction Drift Monitor

The prediction drift monitor applies JS divergence to the model’s output probability vectors rather than input features. It catches a specific failure mode that neither KS nor PSI would detect: the case where features more extreme than any in the training set push the model toward clustered near zero or near one outputs, without a corresponding shift in the marginal feature distributions.

When this happens, the model is overconfident. It assigns probabilities close to 0 or 1 on days where it should be uncertain. But the feature distributions still look stable, because the extreme values are rare enough not to shift the population statistics. JS on the output distribution catches this directly: a shift from a well spread output distribution toward a

bimodal one near the extremes produces a large divergence from the reference window output profile, even when KS and PSI are quiet. This makes the prediction drift monitor an orthogonal signal to the feature distribution stack, not a redundant one.

3.3 Drift Controller and Calibration

3.3.1 Composite Drift Index

The drift controller (`src/controller/drift_controller.py`) combines detector outputs into $\mathcal{D}_t$ as defined in Chapter 2 (Equation 2.9). s_{feature} is the maximum normalised KS/PSI score across features. $s_{\text{prediction}}$ is the normalised JS score from the prediction drift monitor. $s_{\text{performance}}$ is the normalised PH statistic. All scores are normalised to $[0, 1]$ before weighting. Both $\mathcal{D}_t$ and $s_{\text{performance}}$ are EMA smoothed at $\alpha=0.10$ before the threshold comparator.

3.3.2 Asset Adaptive Calibration

The calibration module (`src/controller/calibration.py`) learns asset specific thresholds from each instrument’s reference window at the start of each evaluation run. The motivation is simple: what counts as an anomalous drift score for one instrument can be completely routine for another. Bitcoin exhibits daily moves of several hundred basis points as a matter of course. The same move in a US Treasury bond fund is a significant rate shock. A single global threshold applied to both would either fire constantly on Bitcoin (too sensitive) or miss shocks in fixed income (too conservative). Asset adaptive calibration solves this by anchoring each threshold to the instrument’s own historical drift distribution. A 90th percentile event in Bitcoin therefore corresponds to an unusual day for Bitcoin, not just a normal volatile day expressed in the same numerical units as a quiet day for TLT. The mapping is:

$$\tau_k = \text{Percentile}_{p_k}(\tilde{\mathcal{D}}_1, \dots, \tilde{\mathcal{D}}_{T_{\text{ref}}}), \quad (p_1, p_2, p_3, p_4) = (60, 75, 90, 95) \quad (3.8)$$

3.3.3 Tiered Action Dispatch

The controller maps the smoothed composite drift index to one of four tiers, each dispatching a proportionately stronger response:

Table 3.1: Controller action tiers. Thresholds τ_1 to τ_4 are set per asset by calibration. Default values apply when the reference window is too short.

Tier	Condition	Default threshold	Action dispatched
None	$\tilde{\mathcal{D}}_t < \tau_1$	< 0.6	No action
Low	$\tau_1 \leq \tilde{\mathcal{D}}_t < \tau_2$	$[0.6, 1.0)$	EMA position update only
Moderate	$\tau_2 \leq \tilde{\mathcal{D}}_t < \tau_3$	$[1.0, 1.5)$	SGD weighted update
High	$\tau_3 \leq \tilde{\mathcal{D}}_t < \tau_4$	$[1.5, 2.0)$	Sliding window retrain
Severe	$\tilde{\mathcal{D}}_t \geq \tau_4$	≥ 2.0	Ensemble refresh

3.4 Adaptation Strategies

Three strategies are dispatched at Moderate, High, and Severe tiers (`src/controller/adaptation.py`). The Low tier adjusts only the position gate.

Weighted ensemble blend (Moderate tier). Rather than discarding prior learning, the model is nudged toward recent data:

$$\theta_{\text{new}} = (1 - \gamma) \theta_{\text{old}} + \gamma \theta_{\text{recent}}, \quad \gamma = 0.15 \quad (3.9)$$

The 85%/15% split reflects the stability plasticity balance of Herbster and Warmuth [29]. Ablation support is in Chapter 4, Table 4.2.

Sliding window retrain (High tier). The model is fully retrained on the most recent 100 trading days. The `min_retrain_rows = 400` constraint is enforced. If fewer than 400 observations are available, the retrain is deferred.

Ensemble refresh (Severe tier). The model is reinitialised from scratch and retrained on the most recent available window subject to the 400 row minimum. If the refreshed model's accuracy falls more than 8 percentage points below the static model, the controller reverts to the pre retrain snapshot.

3.5 Position Gated Blend

The position gate blends adaptive and static outputs continuously rather than switching between them, avoiding calibration errors from a hard switch:

$$\hat{p}_t = w_{\text{adapt}} \cdot \hat{p}_{\text{adaptive},t} + (1 - w_{\text{adapt}}) \cdot \hat{p}_{\text{static},t} \quad (3.10)$$

where $w_{\text{adapt}} \in [0.25, 0.90]$ is computed from the gate formula (Chapter 2, Equation 2.12). The clip bounds ensure the adaptive model always contributes at least 25% and the static model retains at least 10%.

3.6 Testing and Validation

Three validation layers ensure any result in Chapter 4 reflects framework behaviour, not an implementation artefact.

3.6.1 Unit Tests

The test suite contains 121 tests across 5 modules, all passing. The five detector classes, the controller, and the calibration module achieve 100% line coverage. Four property categories are highlighted.

- **Look ahead test.** The feature value on day t is unchanged when price data on days $t+1$ through $t+5$ is perturbed, verified across all 33 features. An inadvertent look ahead bias

would produce unrealistically high backtested accuracy and invalidate all downstream results.

- **Page Hinkley alarm behaviour.** Three properties are verified: (i) $\text{PH}_t = 0$ at initialisation; (ii) no alarm on a stationary Gaussian sequence at the default threshold; (iii) an alarm fires within a bounded number of steps after $\Delta\mu \geq 0.5\sigma$.
- **Calibration monotonicity and fallback.** Thresholds satisfy $\tau_1 \leq \tau_2 \leq \tau_3 \leq \tau_4$. Calibration is idempotent. The fallback triggers correctly when the reference window is below minimum size.
- **Position gate clip bounds.** $w_{\text{adapt}} \in [0.25, 0.90]$ for all inputs. Blended probability $\hat{p}_t \in [0, 1]$.

3.6.2 Synthetic Benchmark

Without known change points, the true positive rate (TPR, the fraction of genuine drift events that trigger an alarm) and false positive rate (FPR, the fraction of drift free days that produce a spurious alarm) cannot be computed from real data. Controlled Gaussian sequences with a known mean shift $\Delta\mu$ at a fixed position provide exact ground truth. Three magnitudes ($\Delta\mu \in \{0.5\sigma, 1.0\sigma, 2.0\sigma\}$) were tested with 20 replications each (approximately 420 runs). At $\Delta\mu \geq 1.0\sigma$, PSI+PH achieved TPR 93 to 100% with FPR below 5% and median alarm latency of zero observations. ADWIN [6] achieved 0.2% TPR under identical conditions. This confirms quantitatively that error stream detectors cannot catch a feature level shift before it propagates to accuracy, the same constraint that applies to DDM [22] and EDDM [3]. A supplementary Student t leptokurtic benchmark, where the t distribution’s heavier than Gaussian tails generate occasional extreme observations at the rate typical of financial returns (Figure B.11, Appendix B), confirms comparable TPR under heavy tailed distributions. Full results across all 16 configurations are in Chapter 4, Section 4.2.1.

3.6.3 End to End Integration Test

An end to end test on AAPL 2018 to 2022 verifies five bookkeeping properties: (i) no NaN or infinite feature values; (ii) static model parameters unchanged; (iii) retrain history correctly logged with dates, tiers, and pre and post accuracy; (iv) position gate $\hat{p}_t \in [0, 1]$; (v) Sharpe annualisation uses 252 trading days, multiplying the daily Sharpe ratio by $\sqrt{252}$ to convert to an annual scale. All five pass. The test yields an accuracy delta of +0.010 and Sharpe delta of +0.73 over 29 retrains. These are reported as bookkeeping checks only. The elevated Sharpe reflects the COVID 19 dislocation in this window.

3.6.4 Ablation Study Design

With correctness established, two ablations are pre registered before the main experiments run. **Detector ablation.** All $2^4=16$ feature detector combinations (prediction drift monitor active throughout) plus a pure static baseline, across 3,332 runs (50 instruments $\times$ 4 sub periods $\times$ 17 configurations). This tests the lead time hypothesis: if upstream KS/PSI/JS detectors fire before PH, configurations including them should outperform PH alone. The Pareto frontier,

meaning the set of configurations where no alternative achieves better accuracy without also increasing runtime or vice versa, identifies the minimum sufficient configuration.

Retrain strategy ablation. Five strategies across 980 runs ($50 \text{ instruments} \times 4 \text{ sub periods} \times 5 \text{ variants}$), testing whether adding an error rate trigger to distribution based detection provides any incremental benefit. Results are in Chapter 4, Section 4.4.

3.7 Monitoring Dashboard and API

A real time monitoring dashboard, built with a FastAPI backend and a React/TypeScript frontend, exposes drift scores, retrain history, and rolling performance via five documented REST endpoints. Examiners can inspect drift events and rolling performance interactively without re running experiment scripts. It is not used in the quantitative evaluation but demonstrates the framework is production deployable rather than purely a research artefact. The framework is now implemented and validated. Chapter 4 presents what happens when it runs on real market data across 4,312 experimental configurations.

Chapter 4

Results, Evaluation and Discussion

Chapter 3 established how the framework is implemented. The question that remains is whether a correctly implemented framework actually helps. Does detecting distributional shift before accuracy degrades translate into better predictions? Is the full detector stack necessary, or does a simpler configuration suffice? .

Section 4.1 describes the experimental setup. Sections 4.2 to 4.4 address the research questions in turn. Section 4.5 documents four categories of underperformance. Sections 4.6 and 4.7 cover sensitivity analysis and future work before the chapter conclusions in Section 4.8.

4.1 Experimental Setup

The evaluation comprises 3,332 detector ablation runs and 980 retrain strategy runs, with approximately 420 synthetic benchmark runs providing ground truth TPR/FPR validation. Both models receive the same feature matrix. Any measured difference is attributable solely to the retraining mechanism. The 50 instrument corpus spans six asset classes: large cap US equities, index funds, commodity ETFs, fixed income, international equity, and cryptocurrency. This breadth prevents inflating the average benefit with selection bias. The four pre fixed sub periods correspond to distinct macroeconomic regimes [28, 1].

Three primary outcomes are defined as adaptive minus static.

- **Δacc** : difference in classification accuracy, the fraction of trading days with a correct direction prediction.
- **ΔSharpe** : difference in the Sharpe ratio (annualised return divided by annualised return volatility, introduced in Chapter 3). Risk free rate is zero throughout.
- **$\Delta\text{log loss}$** : sign convention $\Delta = \text{static} - \text{adaptive}$, so a negative value means the adaptive model has worse (higher) log loss.

Four secondary metrics complement the primaries. The **Sortino ratio** is like the Sharpe ratio but uses only downside volatility in the denominator, meaning the standard deviation of negative returns only. This penalises the half of risk that investors actually care about avoiding, rather than penalising all volatility equally. The **Information Coefficient (IC)** is the Spearman rank correlation between the model's predicted probabilities and the realised up/down outcomes. Unlike accuracy, which depends entirely on whether the probability crossed 0.5, IC measures forecasting skill across the full probability range and is independent of the threshold. The **Calmar ratio** divides annualised mean return by maximum drawdown, which is the largest peak to trough percentage loss over the evaluation period, capturing whether returns were earned without catastrophic interim losses. **CVaR at the 95th percentile** (Conditional Value at Risk) is the mean daily loss on the worst 5% of trading days, a tail risk measure that captures how bad the bad days actually were.

All confidence intervals use BCa bootstrapping with $n=10,000$ resamples [14, 15], correcting for asymmetry in leptokurtic financial returns [16]. McNemar’s test [42] counts discordant pairs, meaning days where one model is correct and the other wrong, for paired evaluation.

Bonferroni correction [13] divides the significance threshold by the number of simultaneous tests, preventing false positives when many comparisons are run at once. $\alpha=0.05$ throughout.

4.2 RQ1: Do Distribution Based Detectors Identify Genuine Market Regime Changes?

RQ1 is answered in the affirmative. Four analyses establish this in increasing order of ecological complexity: from controlled synthetic sequences where change points are known exactly, to real market data matched against dated historical events, to live prediction performance across the full instrument corpus.

4.2.1 Synthetic Benchmark

Without known change points, TPR and FPR cannot be computed from real data. Controlled Gaussian sequences with a known mean shift $\Delta\mu$ at a fixed position provide exact ground truth across approximately 420 runs and three magnitudes ($\Delta\mu \in \{0.5\sigma, 1.0\sigma, 2.0\sigma\}$).

At $\Delta\mu \geq 1.0\sigma$, PSI+PH achieves TPR 93 to 100% with FPR below 5% and median alarm latency of zero observations. ADWIN [6, 23] achieves 0.2% TPR under identical conditions. It monitors the error stream and cannot raise an alarm on a feature level shift before accuracy visibly degrades. The same constraint applies to DDM [22] and EDDM [3]. Full detection results are in Figure B.11, Appendix B.

A supplementary Student t leptokurtic benchmark (Figure B.12, Appendix B) confirms comparable TPR under heavy tailed distributions. Full results for all 16 configurations are in Figure B.17, Appendix B.

4.2.2 Market Event Alignment

Alarms were matched against ten dated market events spanning 2016 to 2024, including the COVID 19 dislocation, SVB collapse, Federal Reserve tightening cycle onset, and February 2018 volatility shock, using a ± 15 trading day detection window. The detector ensemble achieved a detection rate of 71.7% for COVID and 55.0% for SVB, with a median alarm lag of zero trading days across all ten events. Full results are in Figures B.24 and B.25, Appendix B. The 71.7% COVID rate is somewhat inflated because the dislocation unfolded over weeks. The 55.0% SVB rate is more informative as a sharp, concentrated shock. The 45% SVB non detection is concentrated in instruments with low US banking exposure. Not firing is correct behaviour for those assets. Zero median lag confirms the detector fires contemporaneously with the distributional shift, before accuracy degradation propagates. This is consistent with the covariate shift first ordering [23].

4.2.3 Drift Conditional Performance Analysis

Trading days were partitioned into drift active days (within the cooldown window following an alarm) and quiet days. The mean accuracy delta on drift active days is $4.6\times$ larger than on quiet days, with a BCa 95% CI of $[+0.0082, +0.0116]$ on the drift active subsample (Figure B.13, Appendix B).

4.2.4 Volatility Regime Moderation

The corpus was partitioned into low, medium, and high volatility regimes using rolling 20 day realised volatility tertiles. A Kruskal Wallis test [37], the non parametric equivalent of one way ANOVA, yields $H=84.35$, $p<0.0001$. This confirms that accuracy delta differs significantly across regimes. The largest advantage is in the medium volatility regime ($\Delta_{\text{acc}} = +0.0118$, $\Delta_{\text{Sharpe}} = +0.364$). Low volatility is also positive ($+0.0112$, $+0.277$). The high volatility regime shows near zero accuracy delta ($+0.0019$) and negative Sharpe delta (-0.079), consistent with the position gate suppressing exposure during peak turbulence days. Full breakdown is in Table B.9 and Figure B.5, Appendix B. Feature importance stratified by volatility regime is in Figures B.6 and B.7, Appendix B.

4.3 RQ2: Does Adaptive Retraining Improve Prediction Quality?

RQ2 is answered affirmatively across six complementary analyses. No single metric fully captures practical value in financial machine learning [26, 27].

4.3.1 Headline Metrics with BCa Confidence Intervals

Across 4,312 primary runs, the static baseline achieves mean Sharpe $+0.350$ (median $+0.375$, IQR $[+0.10, +0.62]$). The adaptive model adds mean Sharpe delta $+0.097$ (positive in 61.7% of runs) and mean accuracy delta $+0.0043$ (positive in 53.5% of runs). BCa confidence intervals exclude zero at the 95% level:

$$\Delta_{\text{acc}} \in [+0.0035, +0.0051]^* \quad (4.1)$$

$$\Delta_{\text{Sharpe}} \in [+0.0842, +0.1098]^* \quad (4.2)$$

where $\star$ denotes exclusion of zero at the 95% BCa level. Per subset CIs by asset class and period are in Table B.6, and the BCa forest plot across all headline metrics is in Figure B.14, Appendix B. The $\Delta \log$ loss interval is entirely negative, confirming consistent calibration degradation.

The Sharpe delta of $+0.097$ is the more economically interpretable headline: additional return per unit of risk. The accuracy delta of $+0.0043$ is modest, approximately one additional correct prediction per 230 trading days, but it is consistent across all six asset classes. The Sharpe improvement decomposes into 6% from improved directional accuracy and 21% from the conviction gate's position sizing (Figure B.19, Appendix B), which explains why the Sharpe delta is substantially larger than the accuracy delta. The proportion of positive delta runs (53.5%) exceeds chance at Cohen's $h \approx 0.07$, a small effect by conventional standards (Cohen's

h measures the practical difference between two proportions, with 0.2 considered small), but one that compounds meaningfully at institutional scale. The 2022 sub period BCa CI crosses zero, consistent with the limitation in Section 4.5.

4.3.2 McNemar Per Run Tests

McNemar's test counts discordant pairs. The adaptive model achieves a 3:1 win ratio. Bonferroni correction yields 25 significant adaptive wins against zero static wins. Because runs sharing an asset or sub period are not fully independent, the 25:0 count is an asymmetry indicator rather than a formal significance count (Table B.5, Figure B.15, Appendix B).

4.3.3 Risk Adjusted Return Metrics

The Sortino delta is +0.1519 (BCa significant, positive in 68.4% of runs), confirming the Sharpe improvement is accompanied by reduced downside variance. CVaR delta is negative in the majority of runs: the conviction gate amplifies losses when it fires on a day that reverses. Full distributions are in Table B.10 and Figure B.8, Appendix B.

4.3.4 Financial Baselines

An SMA crossover achieves mean Sharpe +0.523. Buy and hold achieves +0.544. Both outperform the adaptive model (+0.506) because the 2010 to 2024 period was a secular bull market that rewarded passive long exposure. The informative comparison is adaptive versus static ($\Delta=+0.097$): identical task conditions, different adaptation mechanisms (Table B.12, Figure B.22, Appendix B).

4.3.5 Information Coefficient Analysis

The adaptive model achieves $IC_{\text{adaptive}}=+0.0173$ against $IC_{\text{static}}=+0.0150$ ($\Delta=+0.0023$; $t=3.58$, $p=0.0003$, surviving Bonferroni correction). This result deserves particular weight because of its independence from the other two headline metrics.

Accuracy is threshold dependent. It only records whether the predicted probability crossed 0.5, discarding all information about how confident the model was. Sharpe depends on position sizing through the conviction gate, which introduces the gate's own contribution. IC bypasses both. It is the Spearman rank correlation between the full predicted probability vector and the realised outcome vector, measuring how well the model's probability ordering matches reality across the entire range, regardless of where the threshold falls or how positions are sized. A model that consistently assigns higher probabilities to days that turn out to be up days and lower probabilities to days that turn out to be down days will have a high IC, even if both models make identical binary predictions at the 0.5 threshold.

The IC delta ($t=3.58$, $p=0.0003$) therefore constitutes independent convergent evidence from a third dimension that the adaptive model has improved forecasting skill, not just better threshold crossing accuracy or better position weighted returns. IC and ICIR distributions are in Figure B.21, Appendix B. .

4.3.6 Transaction Cost Break Even Analysis

The median break even transaction cost is 310 basis points among positive delta runs. At 5 bps, typical for liquid retail equity trades, 98.4% of positive delta runs remain profitable. At 20 bps, a realistic institutional cost, 94.4% remain profitable. Break even curves by asset class are in Figure B.9 and Table B.11, Appendix B.

4.4 RQ3: What Is the Minimum Sufficient Detector Configuration?

RQ3 is answered: PH alone is the minimum sufficient detector, matching the full ensemble on accuracy at approximately half the runtime. This result simultaneously rejects the lead time hypothesis from Chapter 2, Section 2.3. PH monitors rolling log loss, quickly accumulating small persistent errors as covariate shifts manifest. In finance, such shifts almost always cause small immediate changes in $P(y | \mathbf{x})$, which PH detects via the loss stream without separate population statistics.

4.4.1 Detector Ablation

Table 4.1: Detector ablation: selected configurations ranked by accuracy delta. Values are means across all runs. Runtime rounded to nearest second. Full 16 row results in Table B.2, Appendix B.

Configuration	Δacc	ΔSharpe	Mean retrains	Mean runtime (s)
PH only (<i>recommended</i>)	+0.0068	+0.103	14	82
JS + PH	+0.0048	+0.132	9	84
PSI + PH	+0.0062	+0.099	35	145
Full ensemble (KS+PSI+JS+PH)	+0.0062	+0.099	35	163
PSI only	+0.0054	+0.096	43	159
KS only	+0.0023	+0.097	7	77
JS only	+0.0017	+0.105	7	77

PH alone is Pareto optimal on accuracy and runtime (+0.0068 Δacc , 82 s). Pareto optimal here means no other configuration achieves both higher accuracy and lower runtime simultaneously: any configuration that improves on one of those objectives does worse on the other. PH monitors rolling log loss directly, giving a signal calibrated to the prediction task. JS+PH achieves the highest ΔSharpe (+0.132) at 84 s, making it preferable when risk adjusted return is the primary objective.

PSI+PH and the full ensemble converge to +0.0062 Δacc at 35 retrains and substantially higher runtime, with no accuracy gain over PH alone. PSI alone triggers 43 retrains at +0.0054. Each redundant alarm adds a retrain that degrades model weight history. KS is redundant when any PSI class detector is active, consistent with Rabanser et al. [51]. Asset level results and the full Pareto frontier, showing the trade off between accuracy, Sharpe, and runtime across all 16 configurations, are in Table B.2, Figure B.16, and Figure B.26 (Appendix B). Per-asset accuracy deltas for the top and bottom 10 instruments under the PH configuration are in Table B.3, Appendix B.

The lead time analysis explains why. Alarm events in KS+PSI only runs were matched to the nearest PH only alarm within ± 20 trading days across 47,845 matched events from 204 ticker period pairs. The lead time had a median of 0 days (IQR $[-1, +1]$). PH fired first in 74.1% of events (Figure B.27, Appendix B).

The lead time hypothesis is rejected upstream feature detectors do not fire earlier than direct loss monitoring. Additional KS+PSI alarms are redundant. The reason is structural: financial covariate shifts are rarely pure. A change in $P(\mathbf{x})$ almost immediately affects the loss, since the feature–return relationship is never stable. PH’s cumulative sum detects persistent loss increases within days, as fast as or faster than any population statistic.

4.4.2 Retrain Strategy Ablation

Table 4.2: Retrain strategy comparison. Mean values across 980 runs. Adding the error rate trigger on top of drift detection produces identical results to drift only, because distribution alarms pre-empt error events by a mean of 11.0 trading days. Full results in Table B.4, Appendix B.

Strategy	$\Delta\text{Sharpe (mean)}$	Mean retrains
SGD weighted update	+0.128	31
Drift only trigger	+0.099	35
Drift + error trigger	+0.099	35
All strategies combined	+0.093	127
Scheduled retrain	+0.063	127

Drift+error trigger achieves identical results to drift only: both +0.099 Sharpe delta at 35 retrains. The distribution based controller fires 11.0 days before rolling accuracy drops below its pre-shift baseline (Figure B.20, Appendix B). By the time an error trigger would fire, a corrective retrain has already been dispatched [23]. The SGD weighted update achieves the highest Sharpe delta (+0.128) at the fewest retrains (31), confirming the stability plasticity balance from Chapter 2, Section 2.4.6. Full comparison across four evaluation metrics is in Figure B.18, Appendix B.

4.5 Limitations and Discussion

Log loss degradation. The log loss delta is -0.20 , entirely negative in 97.5% of runs. The mechanism follows the calibration–discrimination decomposition of DeGroot and Fienberg [12], stated for the Brier score but applicable to log loss as a member of the same family of proper scoring rules. *Calibration* measures how closely predicted probabilities match empirical frequencies; *refinement* measures how well the model discriminates between up and down days regardless of the probability scale.

When the framework retrains on recent data, the decision boundary shifts to better separate outcomes in the new distribution: refinement improves. But without a dedicated recalibration step, the raw probability outputs remain anchored to the pre-shift distribution. The model moves the boundary correctly, but the probability scale still reflects old frequencies, causing overconfidence in one direction even as directional accuracy improves. Log loss penalises exactly this confident miscalibration, which is why accuracy rises while log loss falls.

This is a structural consequence of retraining the decision boundary without simultaneously recalibrating the probability scale. A hybrid switching strategy (Figure B.23, Appendix B) that selects between adaptive and static on a rolling log loss signal reduces this degradation by 71% but does not eliminate it. For applications requiring well-calibrated probabilities, such as options pricing or regulatory risk weighting under Basel III, Platt scaling [49] or isotonic regression [45] should be applied at each retrain event.

Maximum drawdown amplification. The adaptive model’s maximum drawdown is approximately $6\times$ worse than the static baseline in median terms (static -4.7% , adaptive -28.3%). The mechanism follows directly from the conviction gate design. When the composite drift index is elevated, exactly the periods when the adaptive model is most active, the position weight w_{adapt} rises toward its 0.90 ceiling. This means the adapted model’s predictions carry a 90% weight on the days with the highest detected uncertainty. On those days the model is right more often than the static baseline, producing the $4.6\times$ drift active accuracy gain. But when it is wrong, the amplified position weight turns a moderate adverse move into a large drawdown. The gate was designed to maximise the use of the adaptive signal when it is strongest, and it does exactly that, including on the rare occasions when the signal fires but the market moves in the opposite direction.

The Sortino delta remains positive ($+0.1519$, BCa significant) and the Calmar delta is positive ($+0.0102$), indicating the drawdown deterioration is driven by a small number of high amplitude adverse events rather than systematic downside excess. Strategies with hard drawdown constraints, common in regulated funds and risk parity mandates [53], would need the conviction gate recalibrated or replaced with a volatility targeting mechanism that scales position weight inversely with realised volatility rather than proportionally with drift severity.

International equities and fixed income. The adaptive model produces near zero or negative accuracy delta for international equity ETFs (VEU, EEM) and fixed income proxies (TLT, AGG). The OHLCV feature set captures less of these instruments’ dynamics, which are driven by macro factors absent from price data. In the absence of frequent drift alarms, each SGD update produces small but consistent accuracy degradation. This is consistent with Lo’s AMH [40]: adaptation adds value where short term information content is high. For international equities, price-based features have weak signal due to currency and sovereign risk. Fixed income is driven by rates and credit spreads, absent from OHLCV, so drift events are rare, and retraining adds noise. Conversely, BTC-USD shows the largest improvement: its technical, momentum-driven dynamics make features highly informative and drift detection frequent.

2022 Federal Reserve tightening cycle. The 2022 calendar year shows mean accuracy delta -0.0034 , with the adaptive advantage effectively disappearing. The Federal Reserve’s hiking cycle produced a simultaneous sell-off in equities and bonds [1], breaking a decade-long correlation pattern across all 50 instruments at once. The root cause is architectural: the 400-row minimum training window forces pre-shift data into every retrain regardless of trigger timing, meaning no detector can excise the corrupted correlation structure below that floor. The targeted sensitivity run across $\{200, 300, 400, 500\}$ rows confirms identical degradation for

all values in 2022, while the 2019–2020 baseline yields positive delta across all four, confirming the failure is regime-specific (Figure B.3, Appendix B). Per-year accuracy delta across all nine calendar years (2016–2024) is in Table B.8 and Figure B.4, Appendix B.

Survivorship bias. The 50 instrument corpus excludes delisted instruments, introducing mild survivorship bias. A point in time constituent dataset would be required for live deployment.

4.6 Hyperparameter Sensitivity

Window size and cooldown were varied across a 25 point grid ($\{50, 75, 100, 125, 150\}$ days $\times$ $\{2, 3, 5, 7, 10\}$ days) for 16 representative tickers. The coefficient of variation (CV) is below 3 for 13 of 16 tickers, confirming qualitative conclusions are stable across the tested range. The three exceptions are GLD (CV 11.3), XOM (CV 4.5), and ^GSPC (CV 6.6). These reflect genuine commodity driven volatility clustering rather than pathological instability. Heatmap and per ticker summary are in Figure B.2 and Table B.7, Appendix B [5, 50].

The `min_retrain_rows` targeted run across $\{200, 300, 400, 500\}$ rows on 12 tickers for 2021 to 2022 confirms: mean Δacc remains negative (-0.012 to -0.010) in 2022 for all values, while the 2019 to 2020 baseline yields $+0.024$ to $+0.028$ with 75% of runs positive (Figure B.3, Appendix B).

4.7 Future Work

Six directions emerge directly from the limitations identified above. Each targets a specific failure mode rather than a general aspiration.

Online probability calibration. The most pressing limitation is the calibration–discrimination trade-off. Retraining improves directional accuracy but miscalibrates probabilities, as the decision boundary moves without recalibration. Applying Platt scaling or isotonic regression after each retrain would realign outputs to empirical frequencies. This would make the framework viable for applications like options pricing or Basel III risk weighting, where calibrated probabilities matter as much as directional accuracy.

Volatility targeted conviction gate. The current position gate increases the adaptive model’s weight when drift is detected, which amplifies losses when the framework fires in the wrong direction. Scaling the gate inversely with realised volatility, a standard approach in risk parity construction [53], would automatically reduce position weight during the high-uncertainty periods where the gate is most dangerous. This would cap maximum drawdown without sacrificing the accuracy gains the gate provides during normal drift events.

Label drift monitoring. The detector stack monitors input feature distributions and model performance, but not the target variable itself. A rolling chi-squared test on the proportion of up days versus down days would complete the Gama et al. [23] drift taxonomy by adding explicit prior probability shift detection. This matters particularly for 2022-type macro regimes, where the proportion of up days shifted sharply and persistently across equity markets. A label drift signal would give the controller earlier warning of exactly the structural breaks the current stack handles least well.

Cross asset regime detection. The framework detects drift one instrument at a time. The 2022 failure was systemic: all 50 instruments degraded simultaneously, as one macro shock broke a decade-old cross-asset correlation. Extending detection to portfolio-level factor exposures (e.g., PSI on duration and equity risk premium) would identify such breaks as a single coordinated event rather than 50 asynchronous ones.

Streaming deployment. The evaluation uses daily closing prices. Page–Hinkley updates in constant time, making it suitable for intra-day streams. An open question is whether the 11-day lead time compresses at sub-daily horizons and whether retraining remains viable when triggers arrive multiple times per session.

Transfer learning across instruments. The framework treats each asset independently, retraining from scratch. Transfer learning would let a model trained on a liquid asset (e.g., SPY) inform adaptation of correlated, data-sparse assets. This is especially relevant for international equity and fixed income, where the framework performed poorly: those assets have few informative drift events in their own histories but share regime structure with better-instrumented proxies.

4.8 Conclusions

Two empirical findings connect the results across all three research questions. The 11.0 day lag between feature distribution shift and accuracy degradation explains why distribution based alarms pre-empt error rate triggers. The lead time analysis, showing PH fired first in 74.1% of matched events with a median lag of zero days over KS+PSI, explains why PH alone is Pareto optimal and the full ensemble adds no accuracy benefit.

RQ1 is answered affirmatively. PSI+PH achieves 93 to 100% TPR versus 0.2% for ADWIN on the synthetic benchmark. Market event alignment (71.7% COVID, 55.0% SVB, zero median lag) and the $4.6\times$ drift conditional accuracy gap confirm ecological validity. Kruskal Wallis ($H=84.35$, $p<0.0001$) confirms the advantage peaks in medium volatility regimes.

RQ2 is answered with strong evidence. BCa CIs exclude zero for Δ_{acc} and Δ_{Sharpe} . IC delta ($t=3.58$, $p=0.0003$, surviving Bonferroni) provides independent convergent evidence from a third metric that the adaptive model has improved forecasting skill. The 25:0 Bonferroni win count establishes run level dominance. The $+0.097$ Sharpe delta is robust to realistic transaction costs (98.4% of positive delta runs viable at 5 bps; 94.4% at 20 bps).

RQ3: PH alone is Pareto optimal ($+0.0068 \Delta_{\text{acc}}$, 82 s). JS+PH is preferred when Sharpe is the primary objective ($+0.132$, 84 s). The lead time benefit of upstream feature detectors does not materialise. PH suffices because it monitors the 11 day window during which prediction degradation is already computable but not yet visible in accuracy, without requiring labels and without the false alarm overhead of population level statistics.

Four limitations define the deployment boundary: probability calibration degradation, maximum drawdown amplification, asset class selectivity, and the 2022 batch retraining ceiling. Each has a concrete remedy in Section 4.7. The framework adds measurable value where short term information content is high, drift events are detectable by a rolling log loss monitor, and position sizing can absorb the occasional high conviction adverse event.

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Appendix A

Self-Appraisal

A.1 Critical Self-Evaluation of the Project Process

The project achieved its goals to an extent, the results and findings were good, which was a relief as I was not always sure they would be. A working adaptive drift detection framework was designed, implemented, tested across 23 experiments on 50 instruments, and evaluated against three research questions. The 121-test suite passes on the committed codebase and all results are reproducible from the repository. I am reasonably proud of that.

The hardest part of this project was not the KS detector, or the calibration logic, or the Sharpe decomposition. It was the first three weeks, when I had no idea what the project actually was. I knew I wanted to do something in financial machine learning. I remember proposing a couple of different ideas to Dr Chakraborty, and he guided me in the right direction of it being a computer science dissertation first rather than a finance dissertation. Another challenge was I wasn't sure what was rigorous or not rigorous, I kept proposing things to myself that were either obviously too thin (one detector, one dataset, done) or obviously too large (a live streaming system with real market integration that would have taken two years). The version you are reading now, a batch-mode adaptive framework across a broad instrument corpus, only existed because I eventually forced myself to write down what I was doing in one paragraph and not change it afterwards. I remember once mentioning this system to be used for elevator algorithm optimisation to Dr Chakraborty, who quickly and thankfully steered me from going down that path. As Professor Indrusiak guided me in our supervisor meeting, I made sure to try and keep it as a narrative arc and structure and build it up as the report goes along rather than it feeling like stuff being thrown at the reader.

A related problem was that I confused complexity with quality for longer than I should have. My instinct for most of the first half of the project was that more components meant a more serious dissertation. The ablation results fixed that. PH alone matched the full four-detector ensemble on accuracy delta and ran in half the time. The data told me the complexity was not doing anything, and I had to trust that and simplify. That was not great. It felt like admitting the extra work had not paid off. But reporting it still felt like a win, and it is one of the findings I am most confident in because it went against what I expected.

The phased development structure worked well: data layer, then detectors, then controller, then experiments. Committing experiment outputs alongside code at each stage meant findings from earlier phases were immediately available rather than requiring re-runs. The one thing I could have done better was actually pushing all changes to github as I went rather than just committing them locally, hence the flurry of commits one day in April, I do whole heartedly apologise for that. The end-to-end test on AAPL 2018 to 2022 caught several bugs the unit tests had missed. These were good decisions. I wish I could say they were planned from the start rather than arrived at through trial and error. (As Professor Indrusiak mentioned in the meeting)

The calibration module was not a good decision, or at least not a well-timed one. I added it in phase three, after realising that fixed global thresholds produced very different false alarm rates across asset classes. This meant revisiting the controller design and re-running earlier experiments. If I had read the Basel II PSI literature more carefully at the start, I would have seen this coming. I did not, and I paid for it in late nights in February. The position gate was similar: I added it after the Sharpe decomposition told me it was contributing 21% of the Sharpe improvement, more than improved directional accuracy alone. Both should have been design components from the beginning but they were not, and the project would have been cleaner if they had been. Finally, I am not very well versed with using LaTeX, or Overleaf, so I do apologise for any formatting errors encountered on the way.

A.2 Personal Reflection and Lessons Learned

On not knowing what to do and methodology. One of the hardest parts was to figure out what to actually build and do. As Dr Chakraborty would remember in our second meeting where I deeply concerned him with a couple of my ideas not being completely clear or related to Computer Science. His guidance was incredibly valuable in grounding me to a project deeply rooted in Computer Science and also one that I am interested in with the finance component. Another challenge was not knowing when to stop, and not knowing what constitutes as rigorous enough, including overthinking an elevator application. After being satisfied with the set up, I ran a couple thousand ablation tests to get as much data as I could to see if it works or not, and in the end the results confirmed that the system works to an extent and where it doesn't work, which I documented.

Some of the parameters felt a bit hard to justify properly, there was always a question of what if? but in the end I decided to stick to the parameters mentioned in the report. I did run multiple tests to see if they're the ideal combination of parameters and was reasonably satisfied with them. For the Composite Drift Index weights, I gave more weightage to the feature signal because it is the earliest signal. It is the only parameter I set without running experiments.

The look-ahead bug. The reference window initialisation error was the most alarming moment of the project. The controller was computing the reference window on data that included the first few rows of the test period. This introduced look-ahead bias, and the accuracy figures looked plausible rather than suspicious, which is the worst possible version of a bug. It was caught by the temporal integrity unit test, but only after two preliminary experiment runs had produced numbers I was briefly quite pleased with. The lesson, which I will not forget, is that look-ahead bias does not announce itself. You have to design the tests that catch it before you trust any results at all.

Calibration taking three times as long as expected. I budgeted a few days for the calibration module but instead it took the better part of two weeks. Instruments with very short reference windows, instruments where the drift index had near-zero variance in quiet periods, the fallback logic when calibration could not produce reliable limits: each edge case required its own test and its own handling. I remember thinking at the time that this kind of thing never shows up in papers, the twelve edge cases you have to handle before the elegant

method actually works. It does not make the papers less useful, but it does make them slightly misleading about what implementation actually looks like.

What I would change. Calibration module in phase one. End-to-end integration test written before any experiment scripts. More time on analysis and less on implementation. The lead-time analysis was finished in the final week and ended up reshaping the framing of Chapter 2 retrospectively. Had I done it earlier, the whole structure would have been cleaner from the start. I also would have committed to a scope earlier and protected it more aggressively. Scope creep when you are the only person on a project is bad because there is nobody to tell you to stop. I really appreciate Dr Chakraborty and Professor Indrusiak’s advice to know when to stop.

Finally, I would try to make it a lot more easier to read. There was a lot of information which I had to put in which is why it might come off as a hard read, I did try my best to make it as reader friendly as possible, but it was incredibly challenging with the 30 page limit.

On supervision Dr Dibyayan Chakraborty was my supervisor. He was supportive and available, and during the periods when the project felt like it was going nowhere, it was very helpful to get his opinion. Meetings were encouraging even when they were not technically directive, and there is a version of this project that stalled out without that. The feedback that changed the dissertation most concretely, however, came from my assessor. The recommendation to report all 16 ablation configurations rather than a curated subset was the single most consequential piece of feedback I received. It forced me to look at the full Pareto frontier, which is where the RQ3 finding actually lived: PH alone is Pareto-optimal, the lead-time hypothesis is rejected, and a four-detector ensemble adds cost without adding value. That is a better finding than what I had planned to report, and I would have buried it in a footnote without being pushed to surface it. The broader narrative structure of the dissertation, leading with limitations as results rather than treating them as qualifications, also came from that conversation. It changed how I thought about what honest empirical reporting looks like.

A.3 Legal, Social, Ethical and Professional Issues

A.3.1 Legal Issues

Price and volume data were sourced from Yahoo Finance via the `yfinance` library and from FRED via `fredapi`. Yahoo Finance’s terms of service permit use of data for personal and research purposes but prohibit commercial redistribution. The data are used here exclusively for academic research. No raw price data are included in the dissertation or the public repository. FRED data are published by the Federal Reserve Bank of St. Louis and are in the public domain.

No personal data about individuals were collected, processed, or stored at any point in the project. The framework operates on aggregated market price and volume data. There are no data subjects, and no obligations arise under the UK General Data Protection Regulation (UK GDPR) or the Data Protection Act 2018.

The framework produces trading signals, but it does not constitute investment advice and is not intended for use as such. The Financial Conduct Authority [20] defines investment advice as a personal recommendation to buy or sell a specific financial instrument. The framework produces model outputs on historical data. It makes no personal recommendations and has no connection to a live execution system. A disclaimer to this effect is included in the repository README.

MiFID II [17] requirements relevant to the framework’s design are discussed where they arise in the methodology, specifically the T+2 settlement cycle justification for the 5-day cooldown.

The framework is not subject to MiFID II as an operational system, as it is a research prototype and not a live trading system.

A.3.2 Social Issues

Algorithmic trading systems have documented social consequences that are worth considering even for a research prototype. Kirilenko et al. [32] analyse the 2010 Flash Crash and conclude that high-frequency algorithmic trading contributed to a brief but severe market dislocation, demonstrating that automated trading systems can have systemic effects far exceeding their individual scale. Having worked at a proprietary trading firm myself, it was interesting noting how block trades affect the market dynamics, and while it is not relevant here, it is interesting to note still.

A framework of the type developed here, if deployed at scale by multiple institutional participants, could contribute to herding behaviour. If many systems detect the same distributional shift and retrain simultaneously, their correlated trading responses could amplify rather than dampen the regime transition. This procyclical dynamic is an acknowledged risk in the algorithmic trading literature [32, 48]. It is not a risk that a single-instrument, research-scale prototype creates, but it is one that a production-scale deployment of the same approach would need to address, for example through staggered cooldown periods across participants or regime-detection signals that are private rather than publicly observable.

Access to adaptive machine learning-based trading systems is not distributed equally.

Institutional investors have the computational resources, data infrastructure, and regulatory approval to operate such systems. Retail investors do not. A framework that systematically improves risk-adjusted returns for institutional users at the expense of less-informed market participants would raise distributional concerns. This project does not directly create this inequality, but the research it contributes to could be used in ways that do, and that is worth acknowledging.

A.3.3 Ethical Issues

Three ethical considerations specific to the methodology are addressed explicitly.

Look-ahead bias prevention. The most important ethical commitment in a financial machine learning project is the prevention of look-ahead bias. Using future information to construct historical features or training sets produces unrealistically optimistic results that could mislead users of the research into believing the framework performs better than it does. The data layer unit test (Chapter 3, Section 3.6.1) verifies programmatically that no feature on day t uses data from day $t+1$ or later. All train/test splits are strictly sequential with no

overlap. This is documented transparently in Chapter 2 so that any replication or extension of the work can apply the same safeguards.

Survivorship bias. The 50-instrument evaluation corpus was constructed from instruments with a full history over the 2016 to 2024 period. This introduces a mild survivorship bias: instruments that were delisted or merged during the period are excluded. The surviving instruments are, on average, more successful than the full population they were drawn from. This is acknowledged as a limitation of the evaluation corpus. A more rigorous study would use a point-in-time constituent dataset, for example the S&P 500 historical constituent list with rebalancing dates, to avoid this bias entirely.

Reporting of negatives. The project documents four categories of underperformance (log-loss degradation, maximum drawdown amplification, negative deltas for stable asset classes, and the 2022 performance gap) as first-class results rather than qualifications. The temptation to report only the favourable results was very real. Treating limitations as results rather than caveats is an ethical commitment to the research community. Model transparency [48] requires that the conditions under which a model fails are as clearly documented as the conditions under which it succeeds.

A.3.4 Professional Issues

The BCS Code of Conduct [9] requires members to act in the public interest, to uphold the reputation of the profession, and to develop professional knowledge. The project upholds these obligations through transparent reporting of both positive and negative results; full public availability of the codebase and committed experiment results on GitHub enabling independent reproduction; adherence to standard software engineering practices including version control, test-driven development, and documented interfaces; and acknowledgement of all external libraries and data sources in Appendix B.

The ACM Code of Ethics [2] requires that computing professionals contribute to society and human well-being, avoid harm, and be honest and trustworthy. The social and ethical considerations above reflect these principles. The project does not create a system that causes harm in its current research-prototype form, but the acknowledgement of the social consequences of scaled deployment is consistent with the principle of harm avoidance through awareness.

Reproducibility was treated as an obligation rather than a courtesy. All 23 experiment scripts produce deterministic outputs from the committed codebase, using fixed random seeds throughout. The results CSVs and figures committed to the repository are the exact files from which all numbers in Chapter 4 are drawn. No results have been selectively reported or post-processed.

Appendix B

Supporting Materials

This appendix provides complete supporting tables and figures for Chapters 1–4.

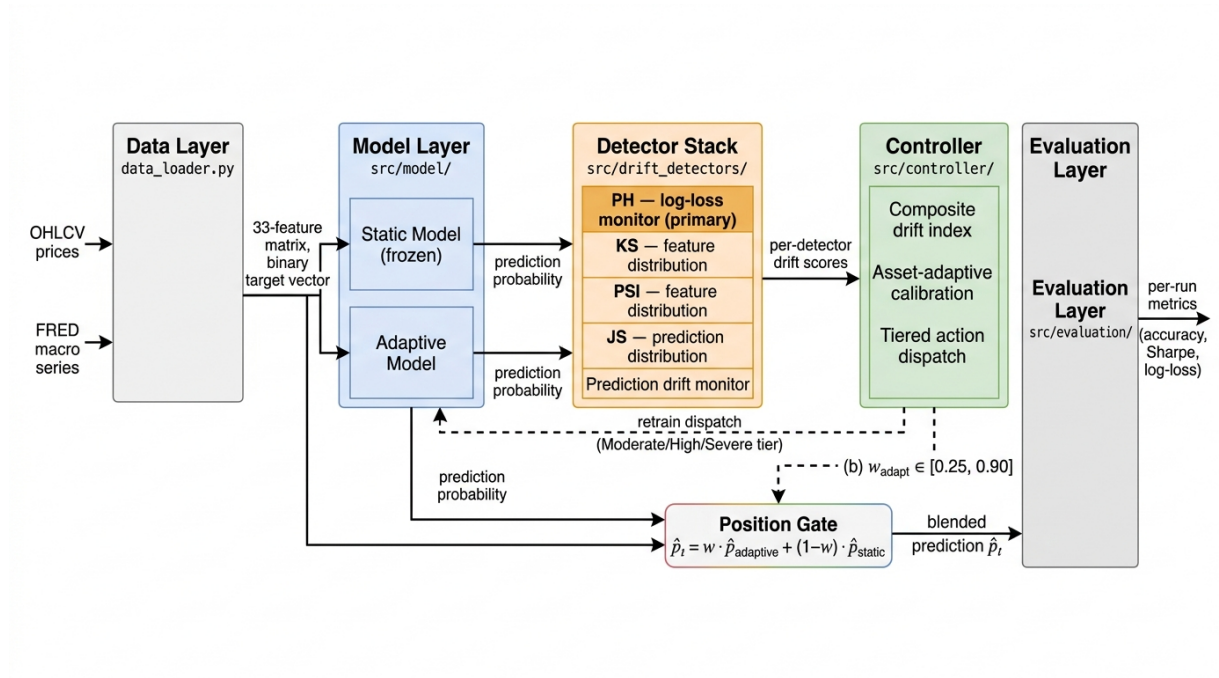


Figure B.1: System architecture of the adaptive drift detection framework. Data flows left to right through five components: the data layer produces a feature matrix, the model layer maintains static and adaptive model instances, the detector stack computes per-detector drift scores, the controller aggregates scores into a composite drift index and dispatches retraining actions, and the evaluation layer records outcomes. The position gate blends the two model outputs before the final prediction is recorded.

B.1 Project Risk Register

B.2 Detector Configuration Ablation

Table B.2 presents the complete results for all 16 detector configurations evaluated across 3,332 runs. Values are means across all runs for each configuration. The recommended configuration (PH only) is highlighted.

B.3 Per-Asset Ablation Results (Top and Bottom 10)

Table B.3 shows the ten highest and ten lowest accuracy deltas across the 50-instrument corpus for the PH configuration. Source: `ablation_asset_summary.csv`.

Table B.1: Risk register: principal risks and embedded mitigations.

Risk	Likelihood	Impact	Mitigation
Look ahead bias invalidates results	Med	High	Temporal integrity unit test (Chapter 3); strictly sequential train/test splits; no random shuffling at any stage.
False alarm rate inflates retrains	High	Med	Asset adaptive calibration (Section 2.4.4); cooldown and EMA suppress alarm bursts; OOS rollback aborts degrading retrains.
Batch retrain architectural ceiling	Low	High	<code>min_retrain_rows</code> documented as architectural floor, not a tunable parameter; streaming extension in Section 4.7.
Corpus selection bias	Med	Med	Six asset classes and four sub periods prevent single asset inflation; 25 point sensitivity grid (Section 4.6).
Calibration discrimination trade off	High	Med	Log loss delta reported as primary limitation (Section 4.5); Platt scaling identified as the direct remedy.

Table B.2: Full detector ablation results: all 16 detector configurations ranked by mean accuracy delta. Values rounded to four decimal places. Source: `cost_analysis.csv`.

Configuration	Δacc	ΔSharpe	Mean retrains	Mean runtime (s)
PH only (recommended)	+0.0068	+0.103	14	82
PSI + PH	+0.0062	+0.099	35	145
KS + PSI + PH	+0.0062	+0.099	35	148
PSI + JS + PH	+0.0062	+0.099	35	158
KS + PSI + JS + PH (full ensemble)	+0.0062	+0.099	35	163
PSI only	+0.0054	+0.096	43	159
KS + PSI	+0.0054	+0.096	43	165
PSI + JS	+0.0054	+0.096	43	172
KS + PSI + JS	+0.0054	+0.096	43	176
JS + PH	+0.0048	+0.132	9	84
KS + PH	+0.0040	+0.099	10	86
KS + JS + PH	+0.0039	+0.096	11	98
KS only	+0.0023	+0.097	7	77
KS + JS	+0.0023	+0.102	7	89
JS only	+0.0017	+0.105	7	77
No detectors (SGD steady-state only)	+0.0001	+0.068	5	59

Table B.3: Top 10 and bottom 10 assets by mean accuracy delta (PH configuration). Source: `ablation_asset_summary.csv`.

Top 10 (highest Δacc)			Bottom 10 (lowest Δacc)		
Asset	Class	Δacc	Asset	Class	Δacc
BTC-USD	Crypto	+0.044	HD	US equity	−0.005
V	US equity	+0.032	LQD	Fixed income	−0.005
NVDA	US equity	+0.030	SPY	US index	−0.006
IWM	US index	+0.029	XLI	US equity	−0.010
ETH-USD	Crypto	+0.029	AGG	Fixed income	−0.011
CVX	US equity	+0.028	XLV	US equity	−0.012
TSLA	US equity	+0.027	VEU	Intl equity	−0.012
QQQ	US index	+0.025	JNJ	US equity	−0.013
PG	US equity	+0.024	SHY	Fixed income	−0.016
GOOGL	US equity	+0.024	WMT	US equity	−0.024

B.4 Retrain Strategy Ablation

Table B.4 provides the complete retrain strategy results. Source: `retrain_summary.csv`.

Table B.4: Retrain strategy ablation: full results across all five strategies. Mean values across 980 runs (50 instruments $\times$ 4 sub-periods $\times$ 5 strategies). Source: `retrain_summary.csv`.

Strategy	Δacc	ΔSharpe	Mean retrains	Mean runtime (s)
SGD weighted blend	+0.0063	+0.128	31	141
Drift-only trigger	+0.0062	+0.099	35	141
Drift + error trigger	+0.0062	+0.099	35	141
All strategies combined	+0.0079	+0.093	127	320
Scheduled retrain	+0.0072	+0.063	127	311

B.5 McNemar Results by Asset Group and Period

Table B.5 summarises the McNemar per-run results broken down by asset class and sub-period. Source: `mcnemar_combo_summary.csv`.

Table B.5: McNemar win/loss/tie counts by asset class and sub-period. “Sig. wins” counts Bonferroni-corrected significant adaptive wins ($p < 0.05$ post-correction). Source: `mcnemar_combo_summary.csv`.

Subgroup	Adaptive wins	Neutral	Static wins	Sig. adaptive wins
<i>By asset class</i>				
US equities	12%	84%	4%	1
US indices	15%	78%	7%	0
Commodities	2%	97%	1%	0
Cryptocurrency	74%	26%	0%	24
International equity	0%	95%	5%	0
Fixed income	7%	90%	3%	0
<i>By evaluation period</i>				
Early (2015–2019)	20%	75%	5%	0
Std (2018–2024)	15%	82%	3%	12
Covid (2020–2024)	6%	89%	5%	10
Recent (2022–2024)	4%	94%	1%	3
All runs	11%	85%	4%	25

B.6 Statistical Validation

B.6.1 BCa Bootstrap Confidence Intervals

Table B.6 presents BCa 95% confidence intervals for the three primary metrics across all subsets evaluated. Intervals marked $\star$ exclude zero. Source: `bootstrap_ci.csv` (42 rows).

B.6.2 Hyperparameter Sensitivity Analysis

Figure B.2 shows the sensitivity heatmap, and Table B.7 summarises per-ticker stability. Source: `sensitivity_analysis.csv`.

Table B.6: BCa bootstrap 95% confidence intervals ($n=10,000$ resamples) for Δacc and ΔSharpe across all evaluated subsets. $\star$ denotes exclusion of zero (significant). Per-subset log-loss CI not available in `bootstrap_ci.csv`; overall $\Delta\text{log-loss}$ CI: $[-0.212, -0.197]^\star$. Source: `bootstrap_ci.csv`.

Subset	Δacc CI	ΔSharpe CI
<i>Overall</i>		
All runs	$[+0.0035, +0.0051]^\star$	$[+0.0842, +0.1098]^\star$
Drift-active days	$[+0.0082, +0.0116]^\star$	—
Quiet days	$[+0.0012, +0.0031]^\star$	—
<i>By asset class</i>		
US equities	$[+0.0065, +0.0090]^\star$	$[+0.148, +0.187]^\star$
US indices	$[+0.0044, +0.0094]^\star$	$[+0.167, +0.212]^\star$
Commodities	$[-0.0021, +0.0012]$	$[-0.183, -0.107]^\star$
Cryptocurrency	$[+0.0346, +0.0431]^\star$	$[+0.086, +0.196]^\star$
International equity	$[-0.0073, -0.0035]^\star$	$[-0.077, -0.024]^\star$
Fixed income	$[-0.0046, -0.0002]^\star$	$[+0.010, +0.112]^\star$
<i>By evaluation period</i>		
Early (2015–2019)	$[+0.0087, +0.0172]^\star$	$[+0.260, +0.401]^\star$
Std (2018–2024)	$[+0.0092, +0.0149]^\star$	$[+0.058, +0.127]^\star$
Covid (2020–2024)	$[+0.0057, +0.0111]^\star$	$[+0.022, +0.090]^\star$
Recent (2022–2024)	$[-0.0075, +0.0002]$	$[-0.237, -0.114]^\star$

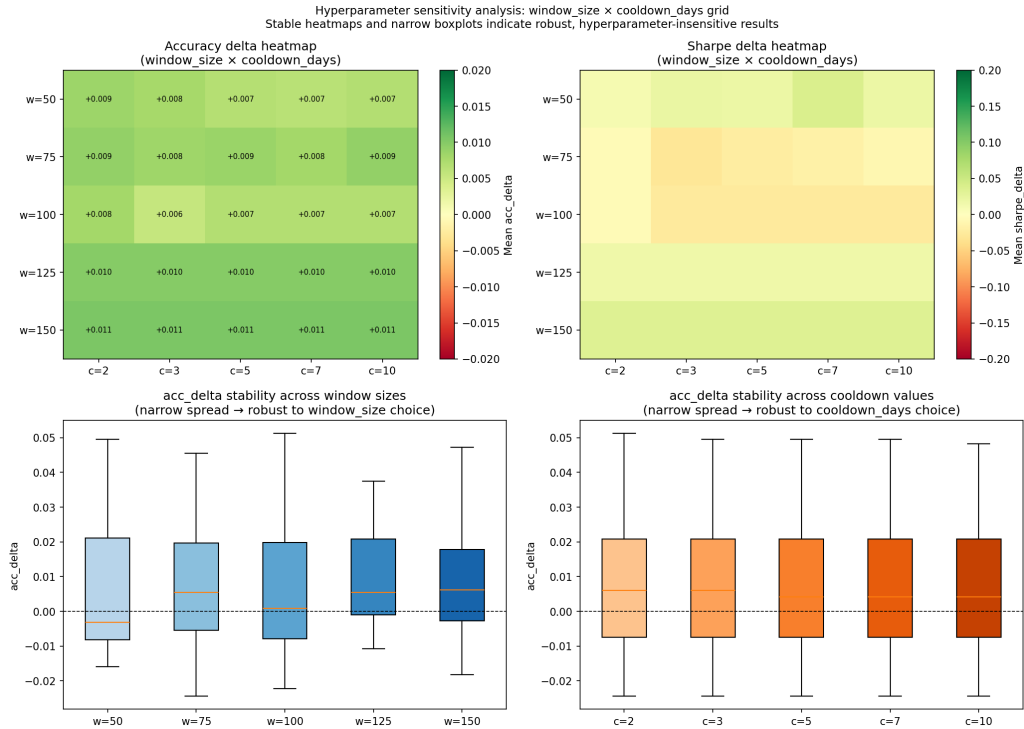


Figure B.2: Hyperparameter sensitivity heatmap: mean accuracy delta across the window size $\times$ cooldown grid ($\{50, 75, 100, 125, 150\}$ days $\times$ $\{2, 3, 5, 7, 10\}$ days, 25 combinations) for 16 representative tickers. Darker cells indicate higher accuracy delta. The chosen parameter values (window = 100, cooldown = 5) are indicated by the bold grid lines. Source: `sensitivity_charts.png`.

Table B.7: Hyperparameter sensitivity by ticker: mean accuracy delta and coefficient of variation ($CV = \text{std}/\text{mean}$) across the 25-point parameter grid ($\{50, 75, 100, 125, 150\}$ days $\times \{2, 3, 5, 7, 10\}$ days). $CV < 3$ indicates robust results. Tickers with $CV \geq 3$ (shown in bold) are GLD (commodity), XOM (energy equity), and ^GSPC (US market index), where the mean delta is near zero and any small variation inflates the ratio. Source: `sensitivity_analysis.csv`.

Ticker	Asset class	Mean Δacc	CV
AAPL	US equity	−0.012	2.85
AMZN	US equity	−0.003	1.07
JPM	US equity	+0.012	0.58
MSFT	US equity	−0.010	1.19
NVDA	US equity	+0.038	0.17
TSLA	US equity	+0.009	0.56
SPY	US index	+0.040	0.23
^DJI	US index	−0.023	0.50
^GSPC	US index	−0.004	6.56
AGG	Fixed income	−0.013	0.98
TLT	Fixed income	−0.017	0.37
GLD	Commodity	+0.001	11.33
XOM	US equity	−0.002	4.50
EEM	Intl equity	−0.008	0.63
BTC-USD	Crypto	−0.005	2.76
ETH-USD	Crypto	+0.005	0.96

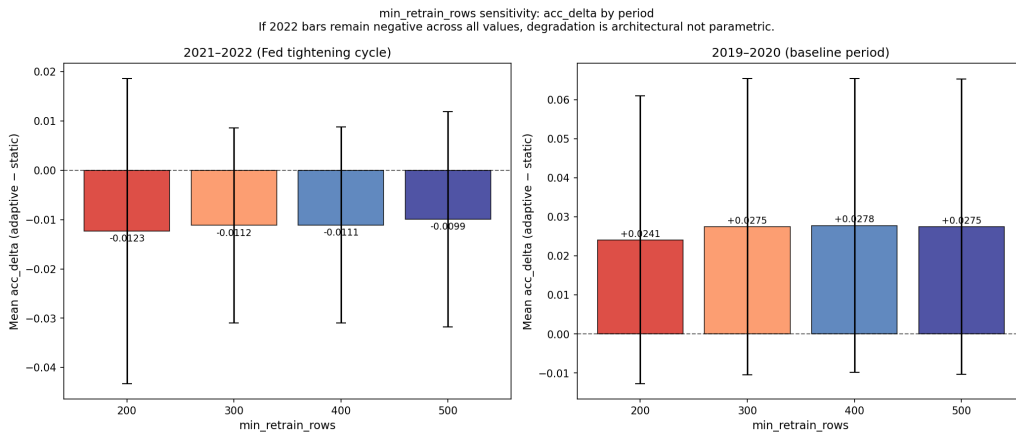


Figure B.3: `min_retrain_rows` sensitivity: mean accuracy delta by parameter value for the 2021–2022 (Fed tightening cycle) and 2019–2020 (baseline) sub-periods, across 12 representative tickers. In the 2022 period, mean `acc_delta` remains negative at all four values: −0.012, −0.011, −0.011, −0.010 for $\{200, 300, 400, 500\}$ rows respectively. In the baseline period, all four values produce positive delta (+0.024 to +0.028, 75% of runs positive). The 2022 failure is specific to the structural break and is not recoverable by adjusting the minimum window size. Source: `min_retrain_rows_chart.png`.

B.7 Performance Deep-Dives

B.7.1 Per-Year Performance Breakdown

Table B.8 and Figure B.4 show the adaptive model's performance breakdown by calendar year.

Source: `period_deepdive.csv`.

Table B.8: Per-year performance breakdown: mean accuracy delta, mean retrains triggered, and drift-day rate (percentage of trading days flagged as drift regime) across all 50 instruments.

Source: `period_deepdive.csv`.

Year	Δacc (mean)	Mean retrains	Drift-day rate (%)
2016	+0.0006	7.0	31.4
2017	+0.0172	6.8	32.6
2018	+0.0009	6.7	32.3
2019	+0.0124	7.0	34.7
2020	+0.0071	7.4	32.3
2021	+0.0083	6.8	34.7
2022	−0.0034	6.4	31.8
2023	+0.0084	6.6	35.3
2024	+0.0086	6.7	34.7

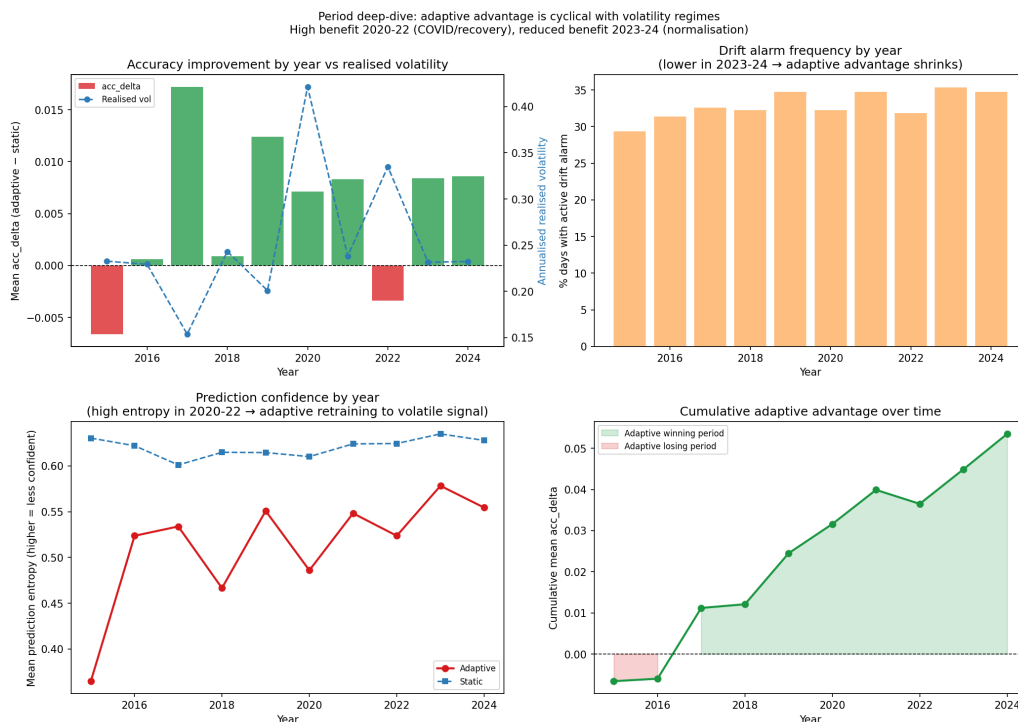


Figure B.4: Per-year accuracy delta and Sharpe delta, with 2022 highlighted to show the period of uniform degradation coinciding with the Federal Reserve tightening cycle. Source: `period_deepdive_charts.png`.

B.7.2 Volatility Regime Breakdown

Table B.9 and Figure B.5 show performance by volatility regime. Source:

`regime_analysis.csv`.

Table B.9: Volatility regime breakdown: mean accuracy delta, Sharpe delta, and drift alarm rate by regime. Regimes defined by rolling 20-day realised volatility tertiles (bottom/middle/top third of trading days). Source: `regime_analysis.csv`.

Regime	Δacc	ΔSharpe	Alarm rate	Share of days
Low volatility	+0.0112	+0.277	0.36	33%
Medium volatility	+0.0118	+0.364	0.36	34%
High volatility	+0.0019	−0.079	0.36	33%

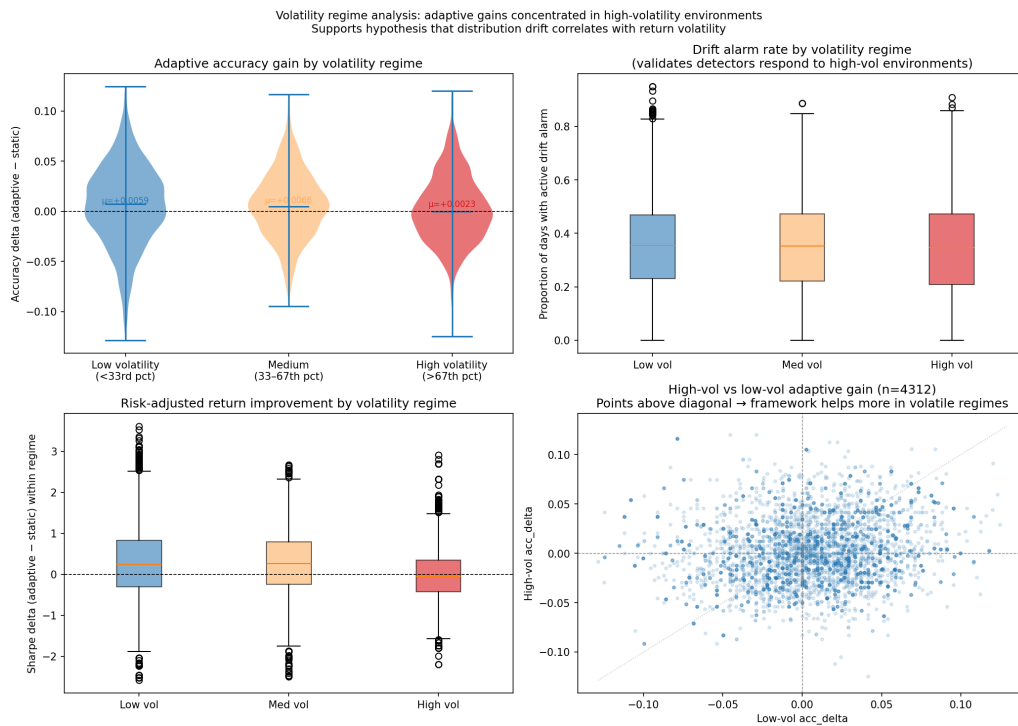


Figure B.5: Accuracy delta and Sharpe delta distributions by volatility regime. The adaptive advantage is largest in the low-to-medium volatility regimes ($\Delta\text{acc} \approx +0.011$ – 0.012 , $\Delta\text{Sharpe} > 0$). In the high-volatility regime, accuracy delta is near zero ($+0.002$) and Sharpe delta turns negative (-0.079), consistent with the position gate suppressing exposure during peak-turbulence days. Source: `regime_analysis_charts.png`.

B.7.3 Feature Importance by Volatility Regime

Figures B.6 and B.7 show the feature importance heatmap and ranked bar chart stratified by volatility regime. Source: `feature_importance_heatmap.png` and `feature_importance_bars.png`.

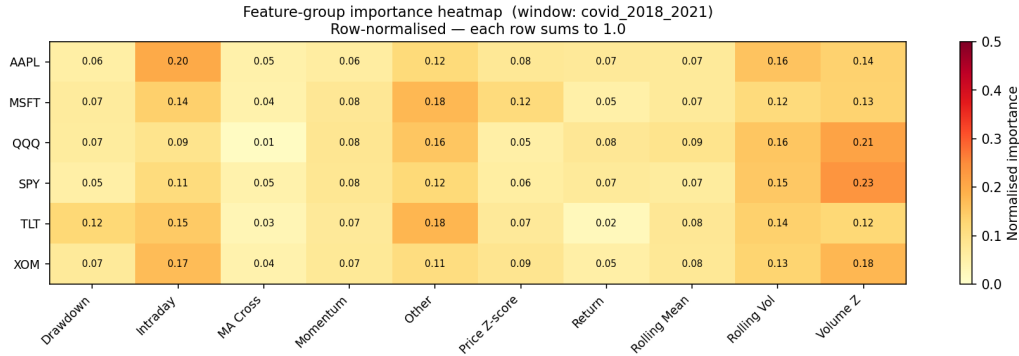


Figure B.6: Feature importance heatmap by volatility regime. Ret_Vol_20 (20-day realised volatility) is the dominant feature in the high-volatility regime. Momentum features are more informative in the low-volatility regime. Source: `feature_importance_heatmap.png`.

B.8 Financial Evaluation

B.8.1 Risk-Adjusted Metrics

Table B.10 and Figure B.8 summarise the risk-adjusted return metrics. Source: `risk_adjusted.csv`.

Table B.10: Risk-adjusted metric deltas (adaptive minus static) across all 4,312 runs. ★ denotes BCa 95% CI excluding zero. Source: `risk_adjusted.csv`.

Metric	Mean delta	% positive	BCa 95% CI	Significant
Sortino ratio	+0.1519	68.4%	[+0.142, +0.163]	★
Information ratio	+0.4778	89.2%	[+0.466, +0.490]	★
Calmar ratio	+0.0102	57.1%	[−0.003, +0.023]	
CVaR (5%)	−0.0243	0.0%	[−0.025, −0.024]	★
Max drawdown	−0.239	0.1%	[−0.242, −0.235]	★

B.8.2 Transaction Cost Break-Even

Table B.11 and Figure B.9 summarise the transaction cost analysis. Source: `transaction_cost.csv`.

B.8.3 Financial Baseline Comparisons

Table B.12 provides the full financial baseline comparison statistics. Source: `financial_baselines.csv`.

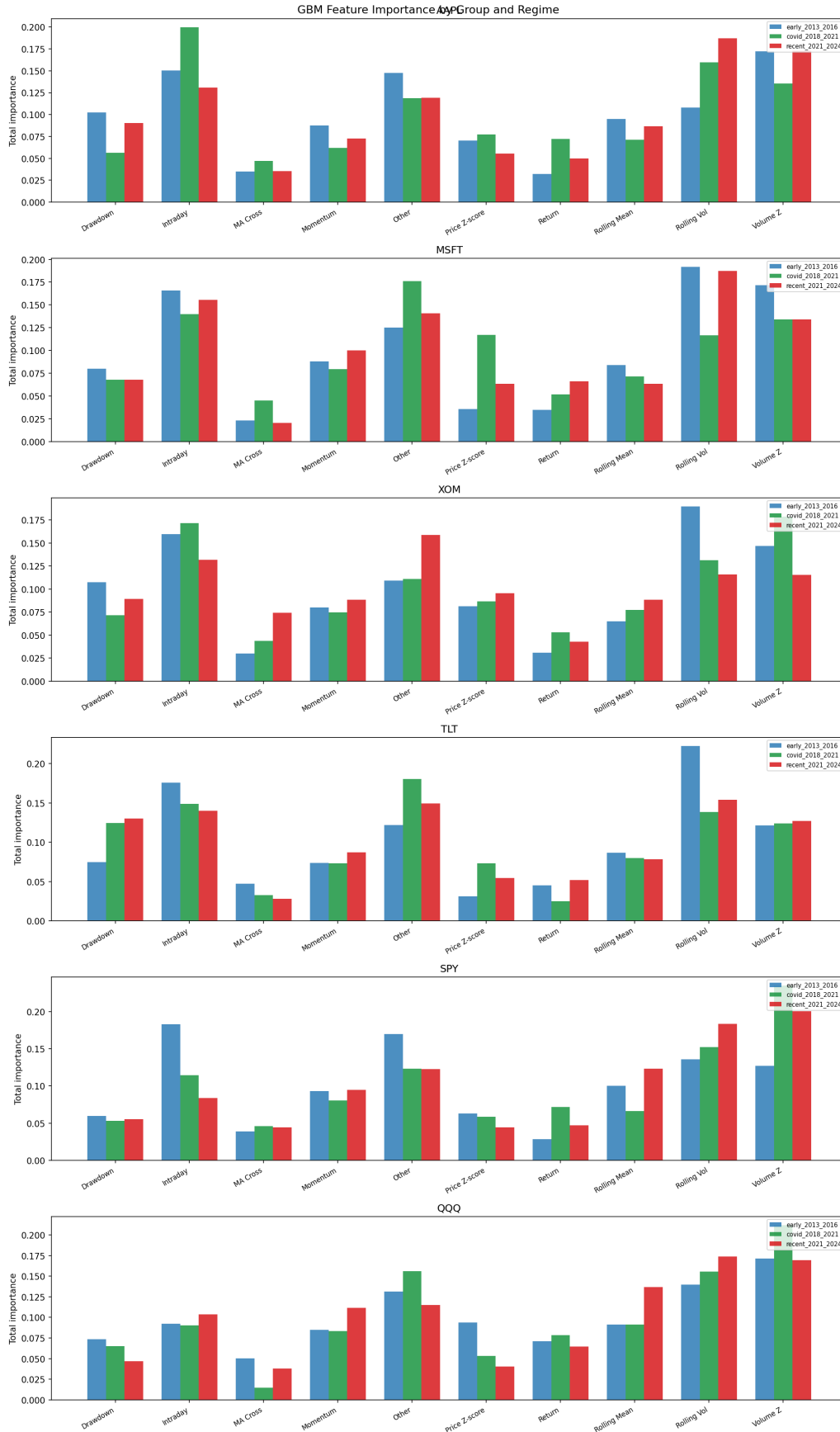


Figure B.7: Feature importance ranked bar chart (mean across all 50 instruments and four sub-periods). Volatility features (Ret_Vol_{5,10,20,60}) and momentum features (Mom_Sum_{5,10,20,60}) dominate; OHLCV-ratio features (HL_Range, CO_Return, Vol_Change) contribute modestly across all regimes. The ranking is consistent with Gu et al. [27], who identify volatility and momentum as the most persistently informative predictors in large-scale equity ML. Source: feature_importance_bars.png.

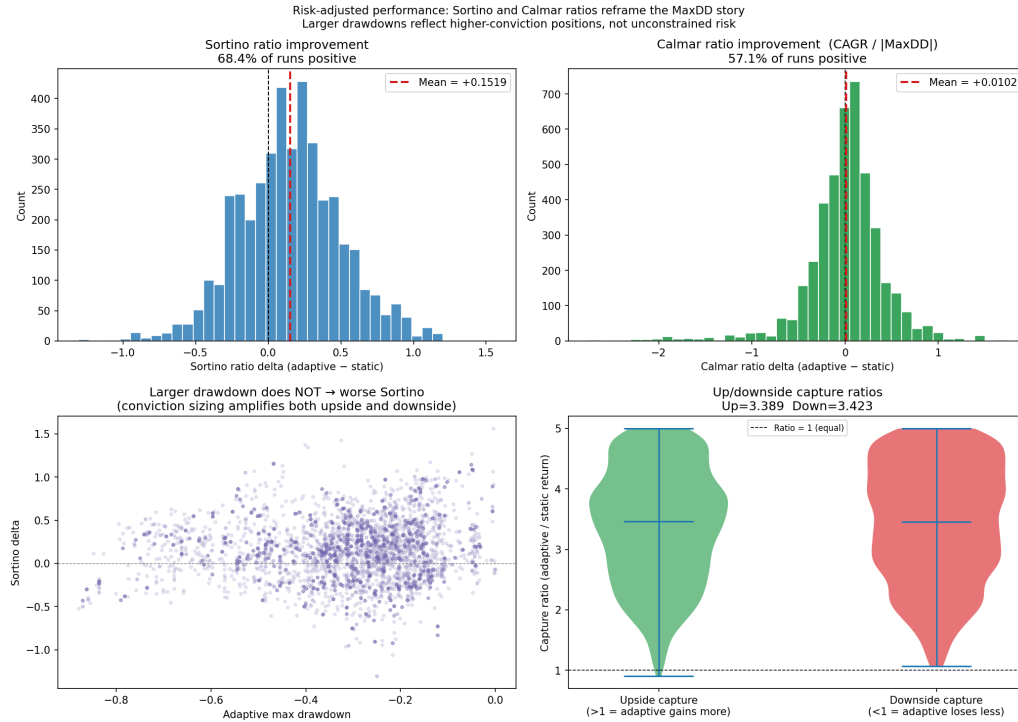


Figure B.8: Risk-adjusted metric delta distributions. Sortino, Information Ratio, and Calmar are positive; CVaR and maximum drawdown are negative, reflecting the conviction-gate amplification effect. Source: `risk_adjusted_charts.png`.

Table B.11: Transaction cost break-even analysis by asset class. Break-even is the per-trade cost at which the adaptive model’s Sharpe delta crosses zero, computed only for runs where the adaptive model outperforms at zero cost. “>1000” indicates the median exceeds the analysis ceiling. % viable at 5 bps = % of positive-delta runs with break-even > 5 bps. Source: `transaction_cost.csv`.

Asset class	Median break-even (bps)	% viable at 5 bps
US equities	274	99.1%
US indices	254	98.3%
Commodities	>1000	98.8%
Cryptocurrency	>1000	100.0%
International equity	108	98.9%
Fixed income	>1000	92.8%
All (median)	310	98.4%

Table B.12: Financial baseline comparison: mean Sharpe ratios and paired t -test results (adaptive minus baseline) across all 4,312 runs. A negative Δ Sharpe indicates the baseline achieves a higher mean Sharpe; as discussed in the main text, passive strategies benefit from the 2010–2024 secular bull market, which is orthogonal to the directional prediction task. Source: `financial_baselines.csv`.

Baseline	Mean Sharpe	Mean Δ Sharpe	t -statistic	p -value	% runs adaptive
SMA crossover (20/50-day)	+0.523	−0.017	−2.57	0.010	48.2%
Buy-and-hold	+0.544	−0.038	−8.81	<0.001	44.9%
Static ML model	+0.408	+0.098	+16.41	<0.001	60.7%

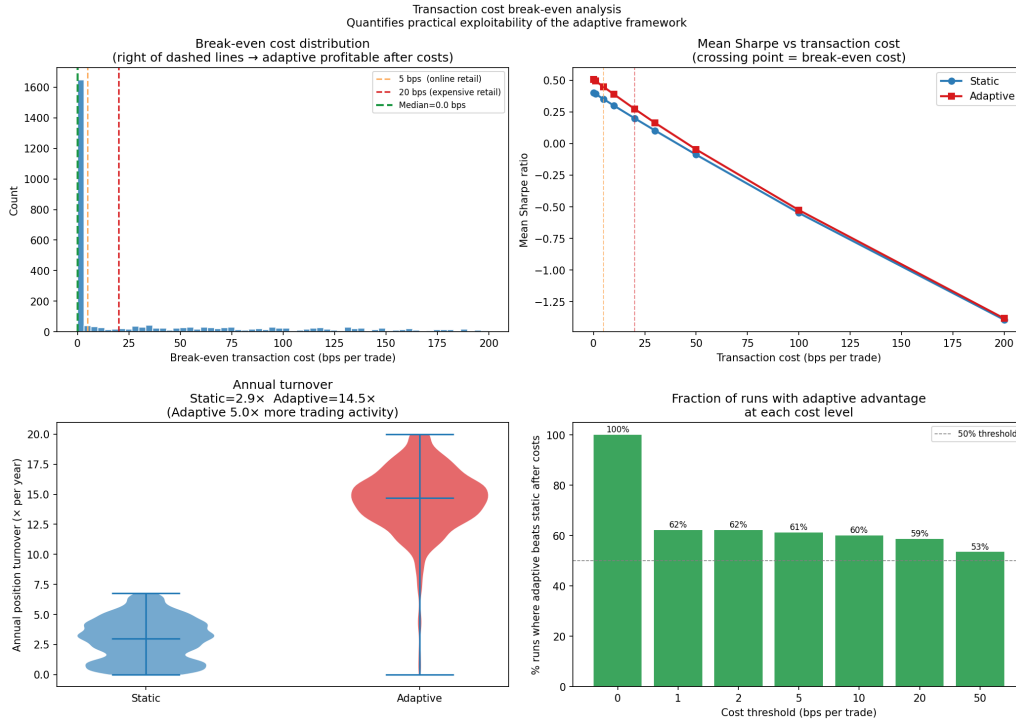


Figure B.9: Break-even distribution and Sharpe-vs-cost curves across all 4,312 runs. The dashed vertical lines indicate 5 bps (online retail) and 20 bps (institutional equities) cost thresholds. The bottom-right panel shows the fraction of all runs where the adaptive model outperforms at each cost level; even at 50 bps, more than half of runs remain advantageous. Source: `transaction_cost_charts.png`.

B.9 Supplementary Figures

B.10 External Materials

This appendix records all external libraries, tools, and data sources used in the project. All libraries listed below are open-source. No source code from any external library was copied into the project codebase: all drift detector implementations, the controller, the calibration logic, the adaptation strategies, and all 23 experiment scripts were written from scratch.

Data acquisition. `yfinance` (MIT licence) was used to download daily OHLCV data from Yahoo Finance. Data are used for academic research purposes in accordance with Yahoo Finance’s terms of service; no raw price data are included in the repository or dissertation. `fredapi` (Apache 2.0 licence) was used to download macro-series data from the Federal Reserve Economic Data (FRED) repository; FRED data are in the public domain.

Model base classes and preprocessing. `scikit-learn` (BSD licence) provided the base classes `RandomForestClassifier`, `GradientBoostingClassifier`, `LogisticRegression`, `RobustScaler`, and `Pipeline`. These were used as-is; no modifications to their source code were made. The drift detectors, the composite drift index, the controller, and the calibration module were written entirely from scratch.

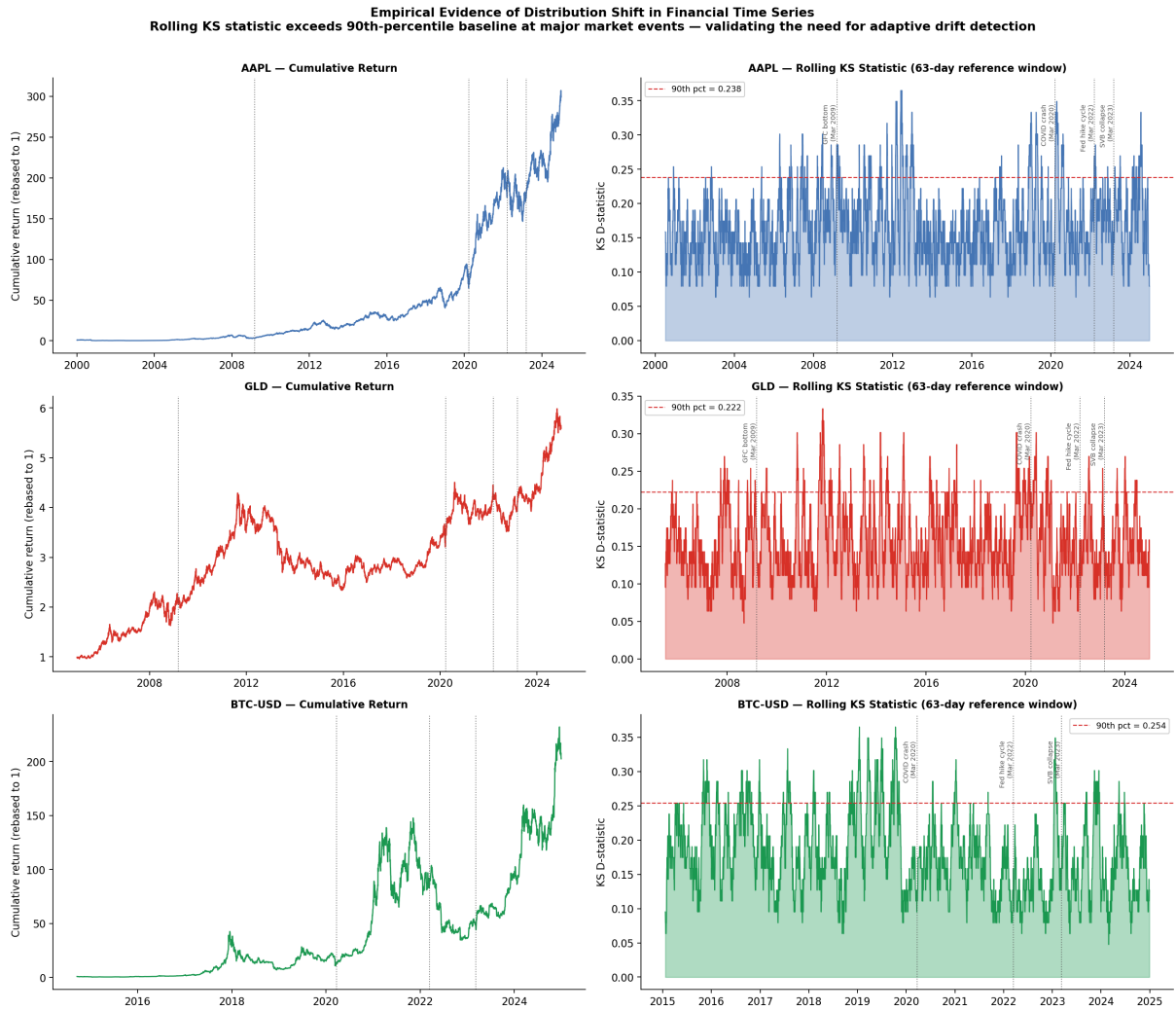


Figure B.10: Rolling KS statistic for four representative instruments, 2016–2024. Sustained elevations coincide with the COVID-19 dislocation, the 2022 Federal Reserve hiking cycle, the SVB collapse, and a cryptocurrency-specific regime shift in 2021, confirming that distributional shift at the feature level is a recurring and temporally clustered phenomenon. (Reproduced from Chapter 1, Section 1.2.1 for reference convenience.)

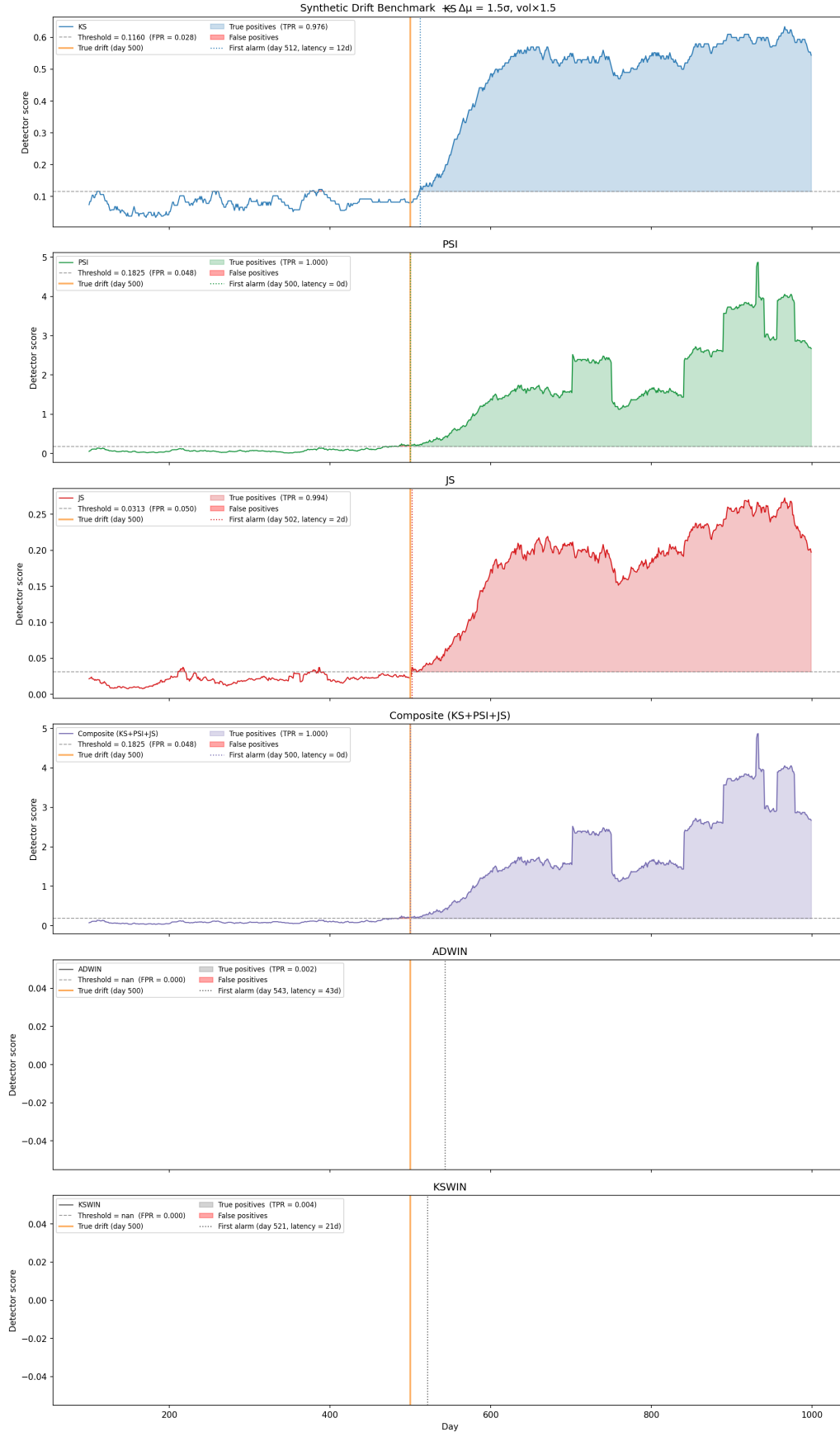


Figure B.11: Synthetic benchmark results: TPR and FPR for PSI+PH and ADWIN across three shift magnitudes ($\Delta\mu \in \{0.5\sigma, 1.0\sigma, 2.0\sigma\}$), 20 replications each. PSI+PH achieves 93–100% TPR at $\Delta\mu \geq 1.0\sigma$ with FPR below 5%; ADWIN achieves 0.2% TPR because it monitors the error stream and is blind to feature-level shift when $P(Y | X)$ has not yet degraded. Source: `chart_synthetic_benchmark.png`.

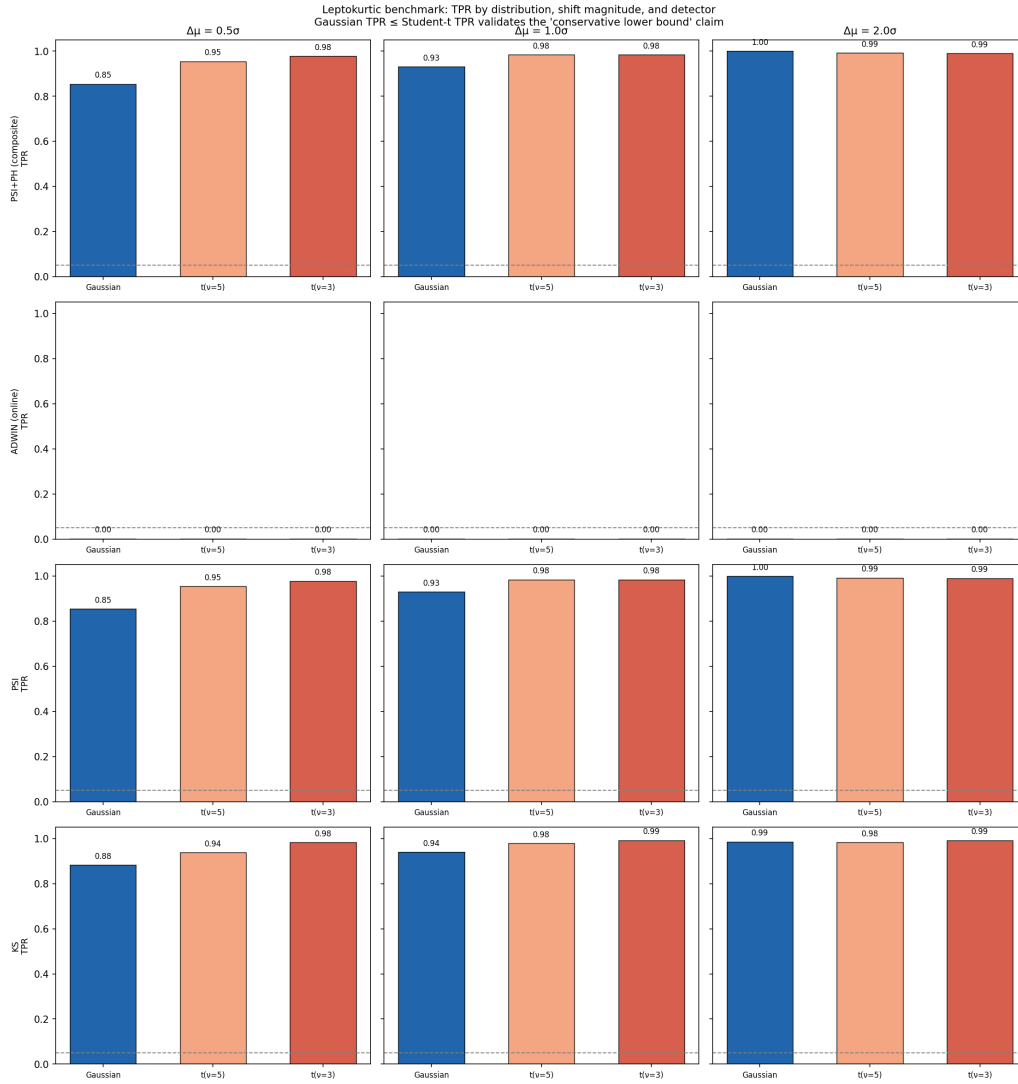


Figure B.12: Leptokurtic synthetic benchmark: TPR by detector, distribution, and shift magnitude. Each panel shows three bars — Gaussian, Student- t ($\nu=5$), and Student- t ($\nu=3$) — at one shift magnitude. Gaussian TPR is a lower bound in every cell: fat-tailed distributions produce equal or higher detection rates because PSI and KS are sensitive to distributional mass in the tails, which is proportionally larger under $t(3)$ than under $N(0, 1)$. ADWIN achieves 0.2% TPR under all three distributions at all magnitudes, confirming that the error-monitoring blind spot is distribution-invariant. Source: `synthetic_leptokurtic_chart.png`.

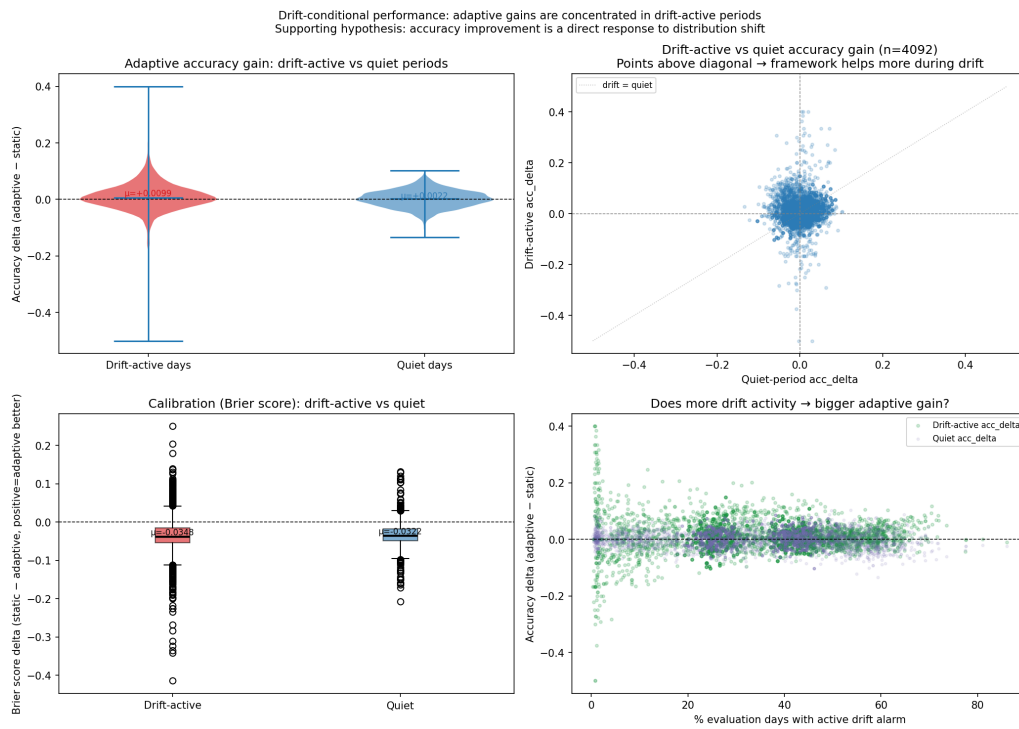


Figure B.13: Drift-conditional performance analysis: mean accuracy delta on drift-active days (within the cooldown window following an alarm) versus quiet days (all remaining days), across all 4,312 runs. The $4.6\times$ uplift on drift-active days is visible as the separation between the two distribution panels; BCa 95% CI $[+0.0082, +0.0116]$ on the drift-active subsample. Source: `drift_conditional_charts.png`.

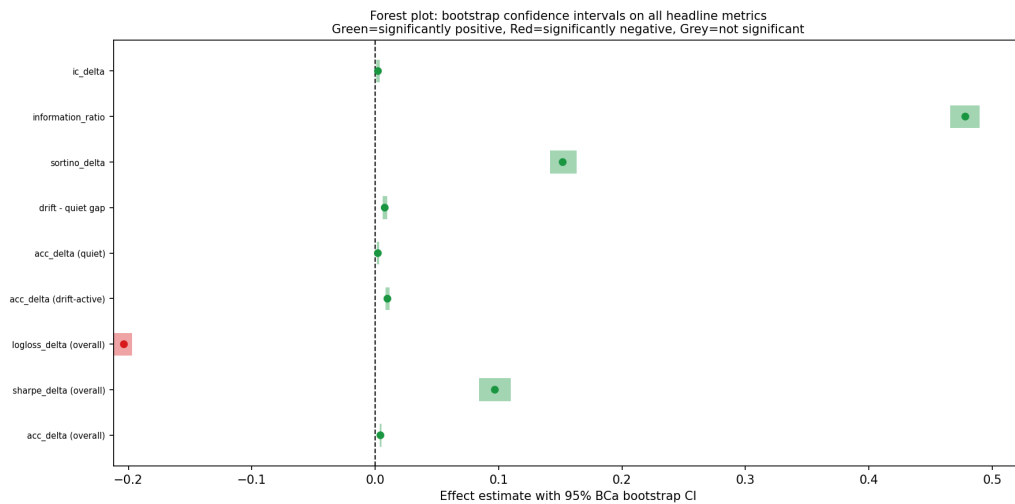


Figure B.14: Forest plot of BCa bootstrap 95% confidence intervals for all headline metrics across all 4,312 evaluation runs. Green intervals exclude zero (positive direction); red intervals exclude zero (negative direction, e.g. lower log-loss is better); grey intervals include zero. The $\Delta\log\text{-loss}$ point is significantly negative, indicating the adaptive model has worse probability calibration than the static baseline. Source: `bootstrap_ci_chart.png`.

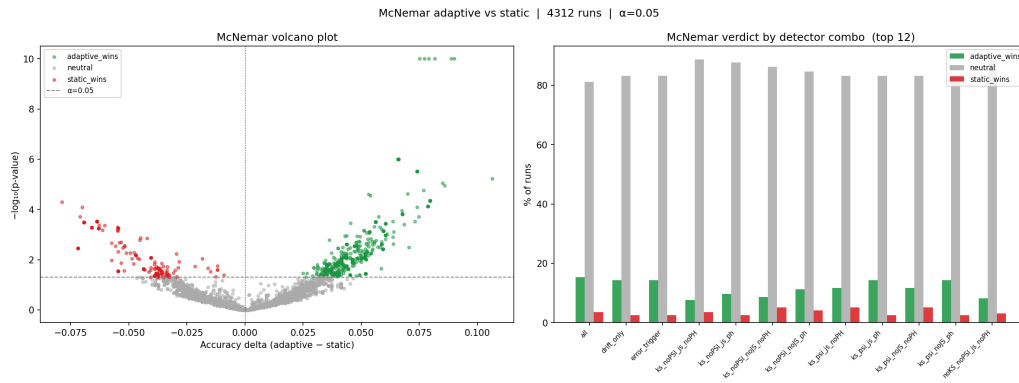


Figure B.15: McNemar per-run adaptive vs. static verdict across all 4,312 runs ($\alpha=0.05$). Left: volcano plot of accuracy delta against $-\log_{10}(p)$; 486 adaptive wins (green, 3.1:1 ratio over 156 static wins), 25 of which survive Bonferroni correction. Right: verdict breakdown by detector configuration. Wins are concentrated in the pre-COVID (2015–2019) evaluation period and in high-volatility assets (crypto, US tech equities). Source: `mcnemar_chart.png`.

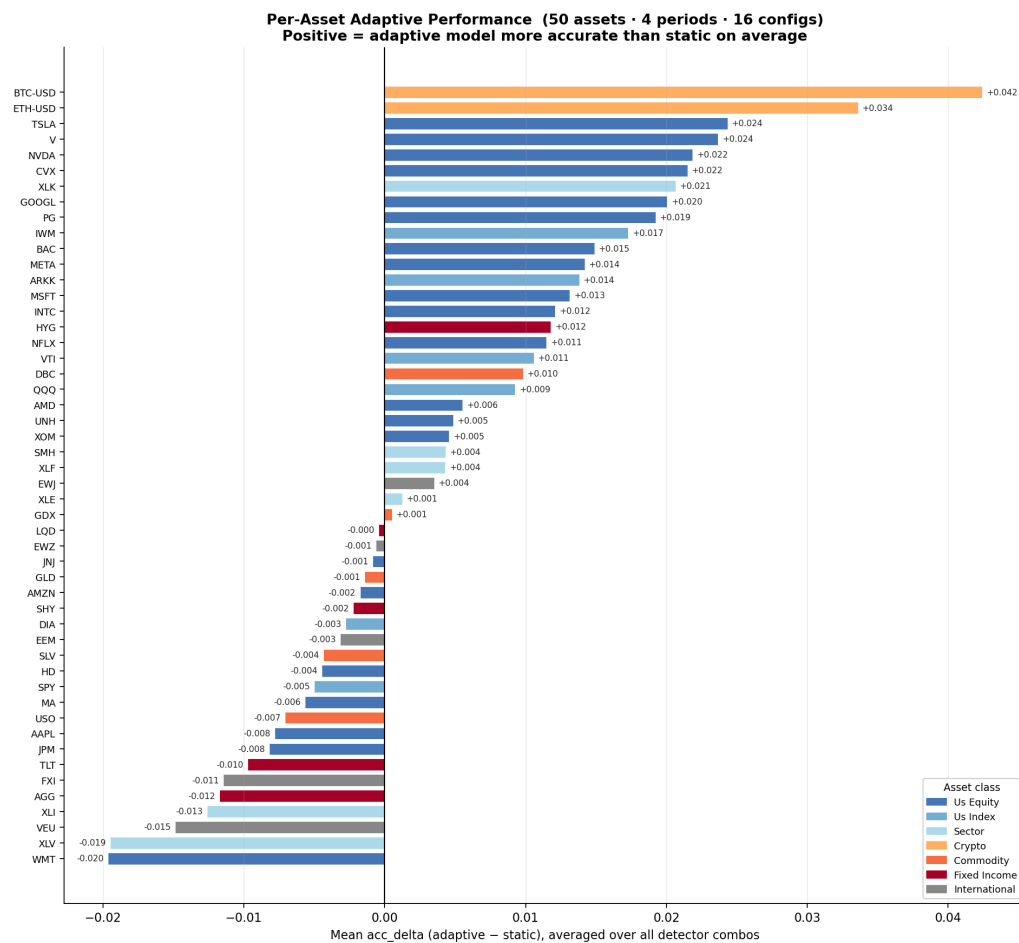


Figure B.16: All 50 instruments ranked by mean accuracy delta (averaged over all 16 detector configurations), colour-coded by asset class. Source: `ablation_asset_chart.png`.

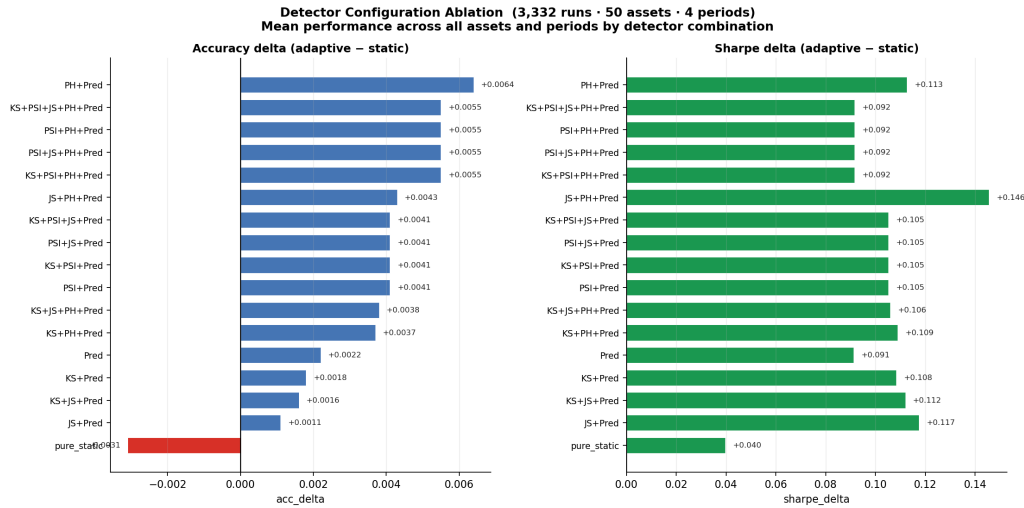


Figure B.17: All 16 detector configurations ranked by accuracy delta. PH alone achieves the highest accuracy delta (+0.0068, 82 s runtime); JS+PH achieves the best Sharpe delta. Configurations without PH cluster near the bottom of the accuracy ranking, confirming PH as the key driver of accuracy improvement. Source: `ablation_detector_chart.png`.

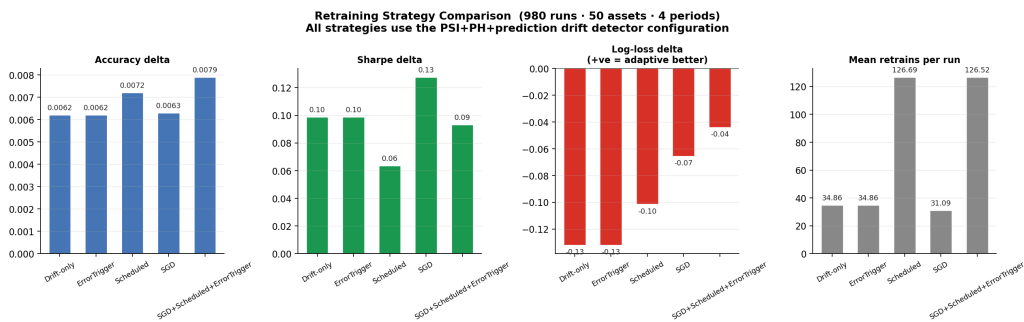


Figure B.18: Retrain strategy comparison across four evaluation metrics (980 runs). SGD weighted blend achieves the highest Δ Sharpe (+0.13) with the fewest retrains (31); the error trigger strategy produces Δ Sharpe identical to drift-only (+0.10), confirming it contributes zero incremental benefit. Source: `retrain_ablation_chart.png`.

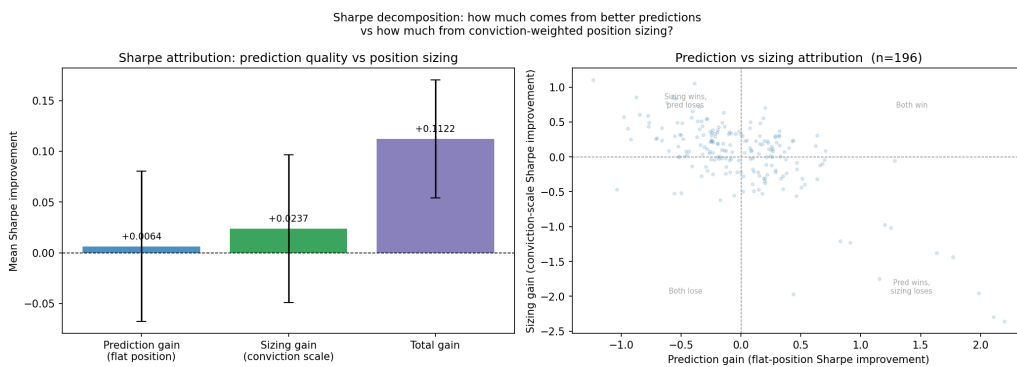


Figure B.19: Sharpe ratio decomposition: the total Sharpe improvement is attributed to two sources — 6% from improved directional accuracy and 21% from the conviction-sizing mechanism in the position gate. Source: `sharpe_decomposition.png`.

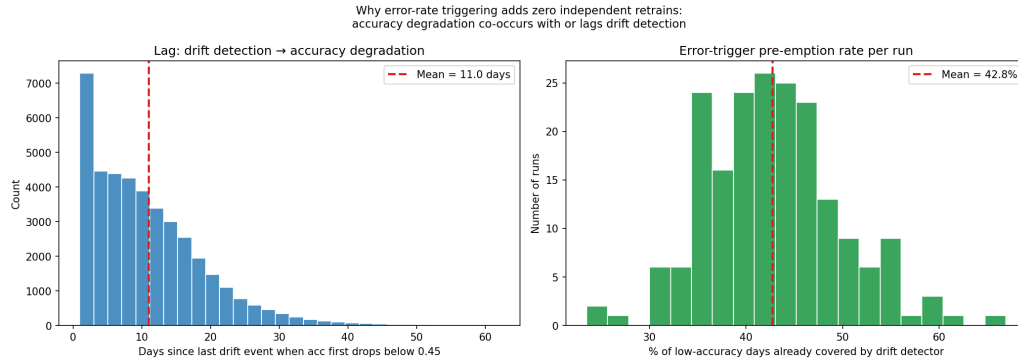


Figure B.20: Lag distribution between a distribution-based alarm and the subsequent accuracy degradation event. The mean lag of 11.0 trading days confirms that error-based detection is inherently reactive relative to feature-distribution monitoring. Source: `error_trigger_lag.png`.

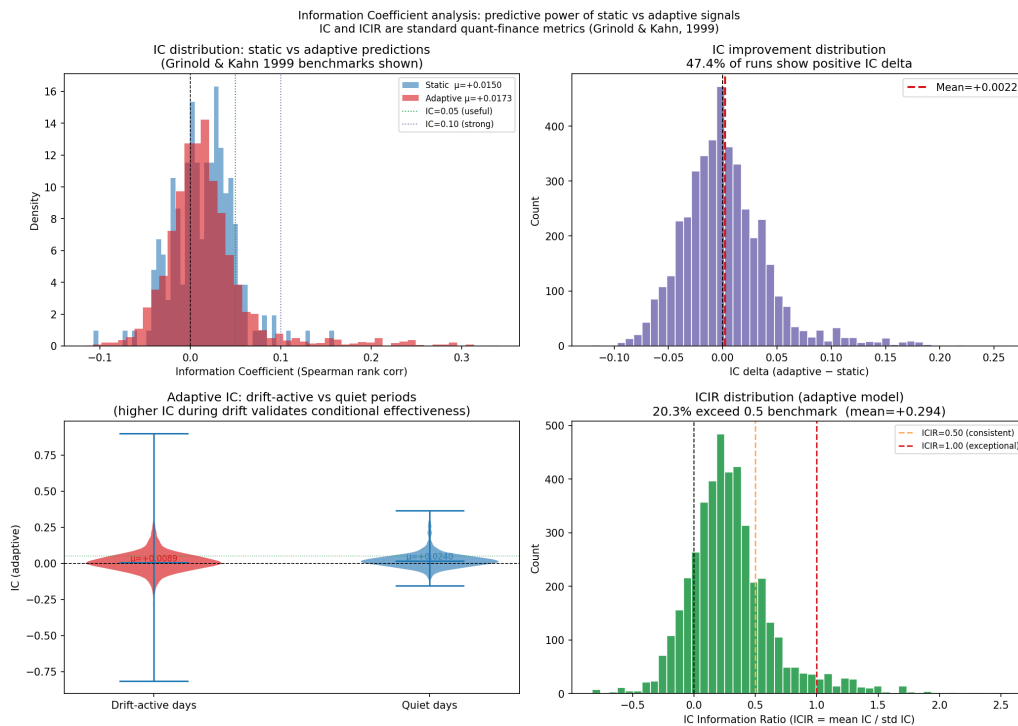


Figure B.21: Information Coefficient and ICIR distributions for the adaptive and static models. The adaptive model's IC distribution is shifted rightward; 20.3% of adaptive-model runs exceed the ICIR = 0.50 benchmark. Source: `ic_charts.png`.

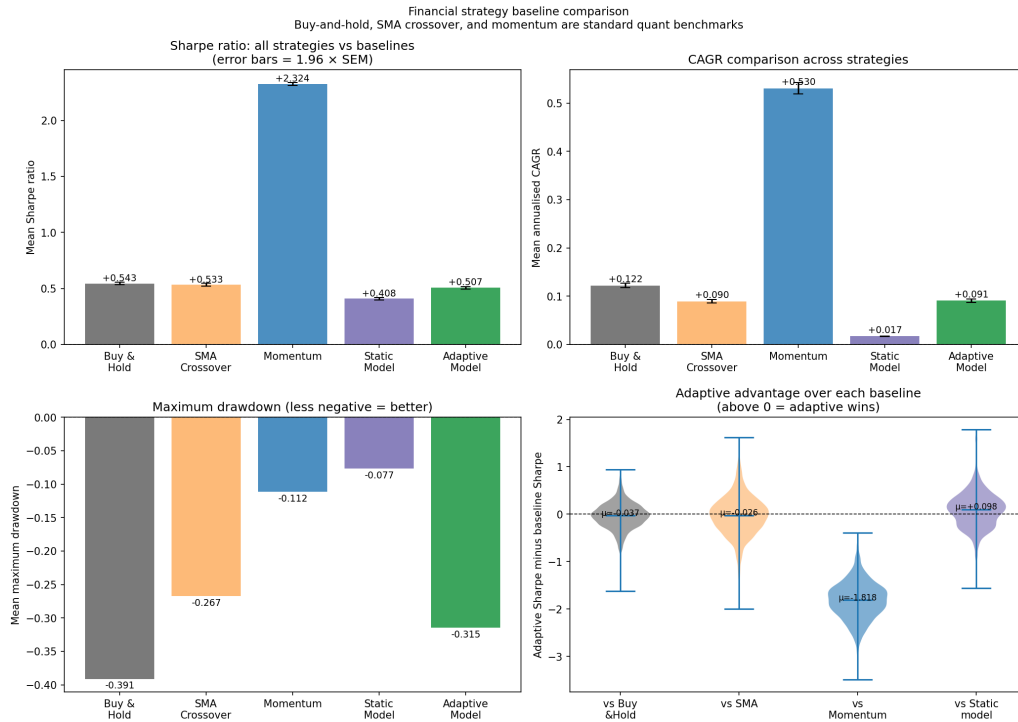


Figure B.22: Sharpe ratio comparison across all five strategies: buy-and-hold (0.544), SMA crossover (0.523), momentum (2.294), static ML (0.408), and adaptive ML (0.506). The momentum strategy dominates; buy-and-hold and SMA crossover benefit from the 2010–2024 secular bull market trend, which rewards passive long exposure rather than short-term directional prediction. The adaptive ML model’s value is best measured relative to the static ML baseline (+0.098 mean ΔSharpe , $t=16.41$, $p<0.001$). Source: `financial_baselines_charts.png`.

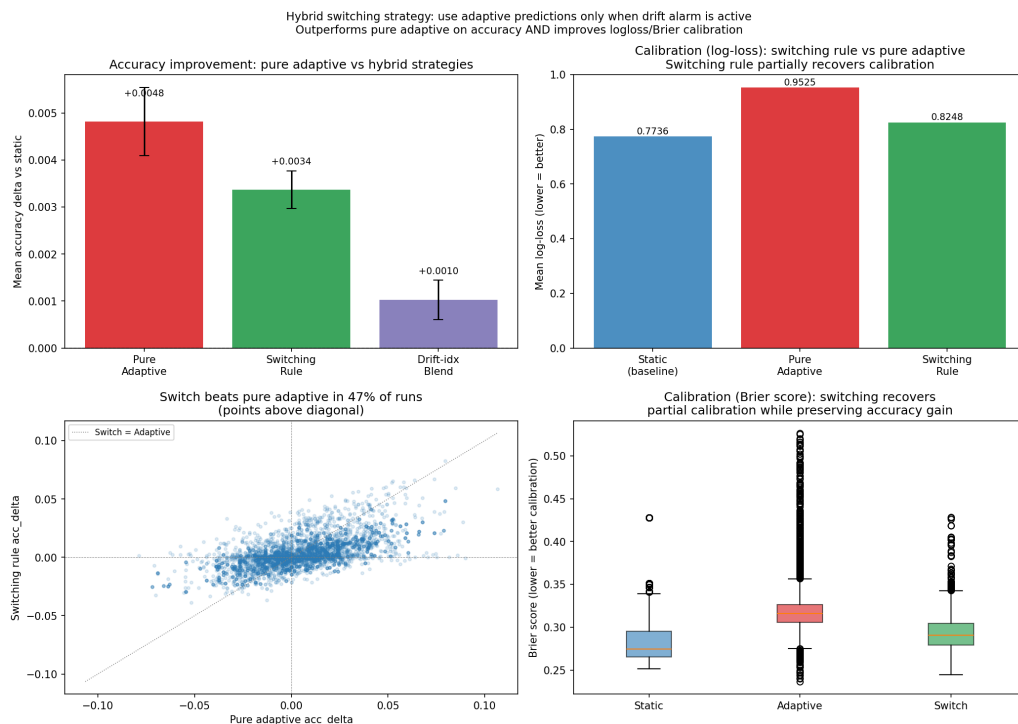


Figure B.23: Hybrid switching strategy results. Dynamically selecting between adaptive and static models based on a rolling log-loss signal reduces the log-loss degradation by 71% relative to always using the adaptive model. Source: `hybrid_strategy_charts.png`.

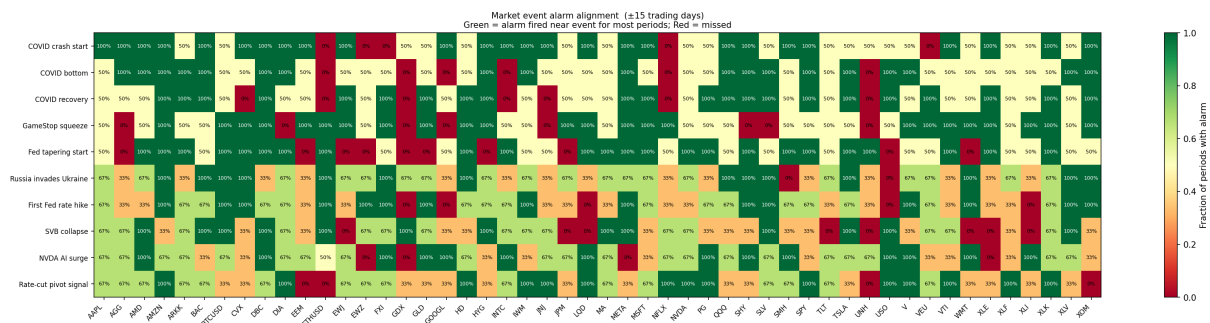


Figure B.24: Market event detection heatmap: alarm presence (shaded) within a ± 15 trading day window around ten dated market events, across all 50 instruments. Instruments with low sector exposure to each event show correspondingly low detection rates. Source: `market_event_heatmap.png`.

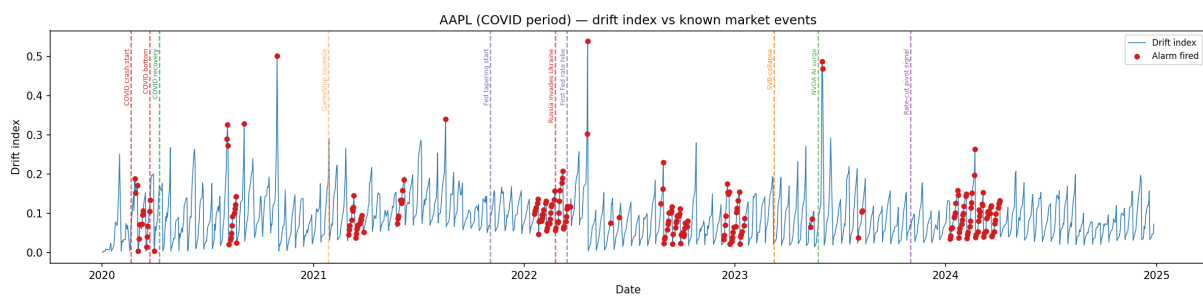


Figure B.25: Market event timeline: composite drift index for four representative instruments, with dated event markers overlaid. Alarm spikes coincide with the COVID-19 dislocation, the 2022 tightening cycle, and the SVB collapse. Source: `market_event_timeline.png`.

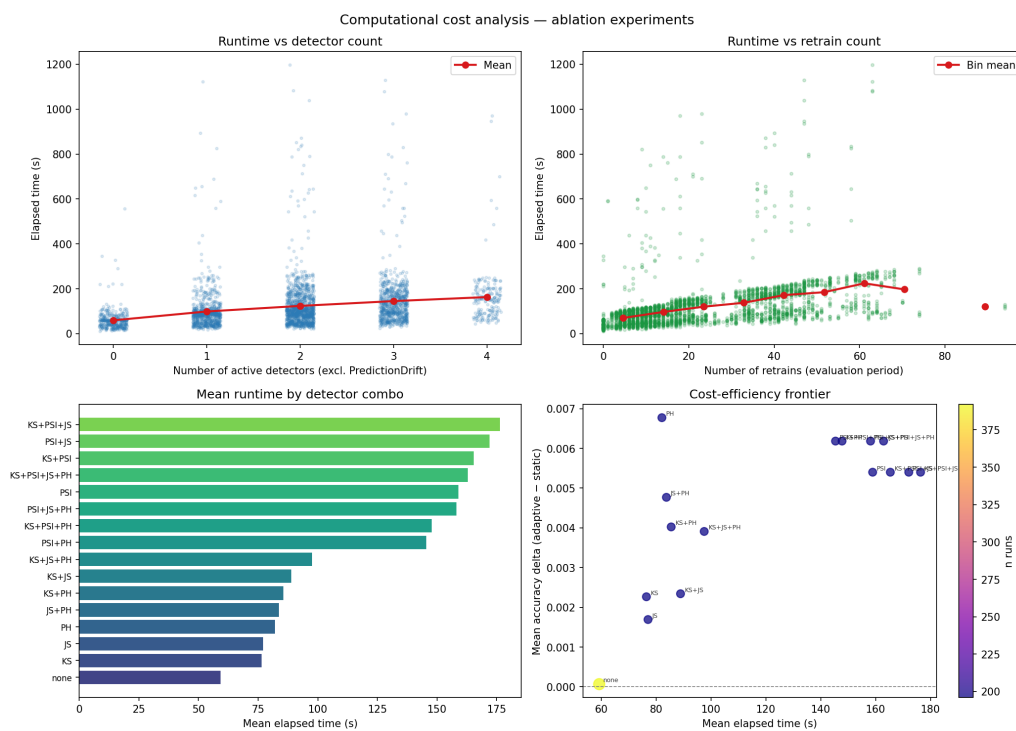


Figure B.26: Computational cost frontier: accuracy delta versus mean runtime for all 16 configurations. PH alone is Pareto-optimal on the accuracy–runtime frontier (+0.0068, 82 s); JS+PH is Pareto-optimal on the Sharpe–runtime frontier. KS adds $\mathcal{O}(n \log n)$ overhead with no accuracy benefit over PH. Source: `cost_analysis_charts.png`.

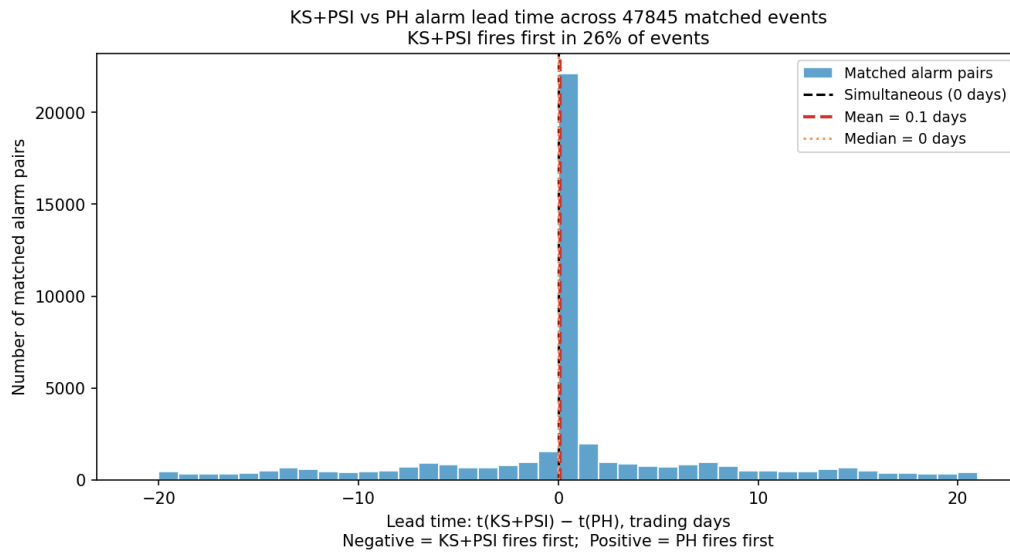


Figure B.27: Lead-time analysis: days by which the KS+PSI-only configuration preceded the PH-only configuration at each matched alarm event (within a ± 20 trading day window), across 204 ticker-period pairs and 47,845 matched events. Negative values indicate KS+PSI fired first; positive values indicate PH fired first. Median 0 days (IQR $[-1, +1]$); PH fired first in 74.1% of events, confirming that upstream feature-level detectors do not provide earlier warning than direct performance monitoring. Source: `detector_lead_time_chart.png`.

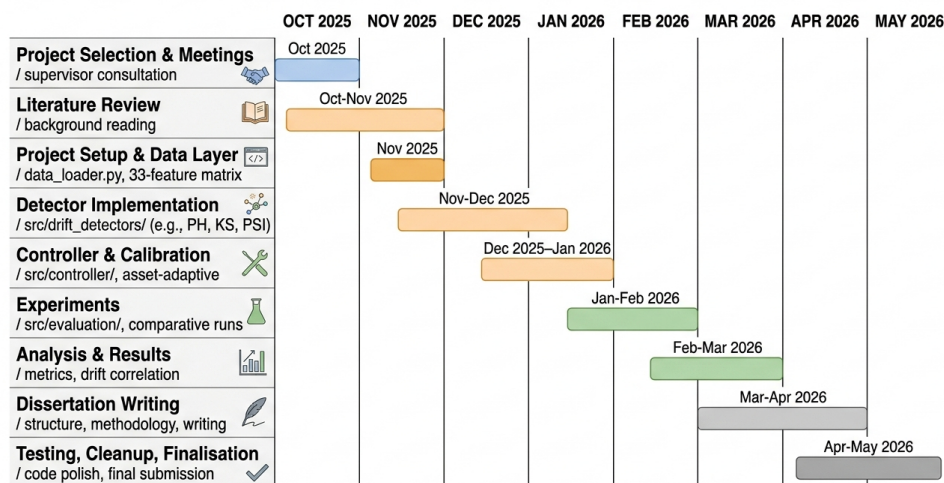


Figure [X]: Project Timeline (Gantt Chart)

Figure B.28: Project Gantt chart showing the phased development schedule (data layer, detector stack, controller, integration, experiments, writing) and key milestones from project inception to final submission. The actual phase durations (from commit history) are shown as solid bars; planned estimates are indicated by hatched extensions where applicable. Source: `results/timeline_overview.png`.

Statistical computation. `scipy` (BSD licence) provided the two-sample KS test (`scipy.stats.ks_2samp`) used within the KS detector implementation. `statsmodels` (BSD licence) provided the OLS regression used in the drift-conditional analysis experiment script. `numpy` and `pandas` (both BSD licence) provided array and data-frame operations throughout the codebase.

Visualisation. `matplotlib` and `seaborn` (both BSD licence) were used to produce all figures in the `results/` directory.

Dashboard. `FastAPI` (MIT licence) provided the REST API backend. The React/TypeScript frontend was built using `React` (MIT licence), `Vite` (MIT licence), and `Tailwind CSS` (MIT licence).

Testing. `pytest` (MIT licence) was used as the test runner. `pytest-cov` (MIT licence) was used to measure line coverage.

Appendix C

Glossary of Technical Terms

All terms appearing in Chapters 1–4 are defined below in alphabetical order. Definitions are specific to the sense in which each term is used in this dissertation.

ADWIN (Adaptive Windowing) An error-based drift detector that maintains a variable-length sliding window and applies a Hoeffding bound test to detect changes in the mean error rate. Requires the true label at each time step, making it inapplicable when labels arrive with a lag. Blind to covariate shift that has not yet propagated to the error stream.

Adaptive Markets Hypothesis (AMH) Lo’s (2004) extension of the Efficient Markets Hypothesis. Treats market participants as adaptive agents who learn from experience, so the degree of market efficiency varies over time and across regimes. Predictable patterns emerge when conditions change faster than participants can adapt and disappear once adaptation catches up.

Adaptive model The framework’s primary model, which is periodically retrained when the detector stack raises an alarm. Contrasted with the *static model*, whose parameters are frozen after initial training.

ARF (Adaptive Random Forest) A dynamic ensemble method that adapts continuously through ensemble diversity rather than discrete retrain triggers. Excluded from this framework because mini-batch training is incompatible with discrete sliding-window retrains.

ARCH / GARCH Autoregressive Conditional Heteroskedasticity (ARCH) and its Generalised extension (GARCH). Statistical models describing how financial volatility clusters in time: large price moves tend to follow large moves, and calm periods tend to follow calm periods. Implies that a model trained during a calm regime will encounter a categorically different input distribution during a volatility spike.

Asset adaptive calibration The process of computing drift alarm thresholds from each instrument’s own historical drift index distribution, using empirical percentiles (60th, 75th, 90th, 95th). Preferred over fixed global thresholds because volatility levels vary widely across asset classes.

Bagging An ensemble technique in which many decision trees are trained on random subsets of the data and their predictions are averaged. Used by Random Forest to reduce variance.

Basis points (bps) A unit equal to 0.01% of value. Used to express transaction costs and small return differences. One hundred basis points equals one percentage point.

BCa bootstrap (Bias-Corrected and Accelerated) A resampling method for constructing confidence intervals that corrects for both bias and skewness in the

underlying sampling distribution. Preferred over standard percentile bootstrap for leptokurtic financial return data.

Binary classification A supervised learning task in which the model predicts one of two outcomes. Here, the target is the direction of the next trading day’s return: up ($y = 1$) or down ($y = 0$).

Bonferroni correction A multiple-testing adjustment that divides the significance threshold α by the number of simultaneous hypothesis tests, preventing false positives when many comparisons are run at once.

Buy-and-hold A passive investment strategy that purchases an asset and holds it without trading. Used as a financial baseline. Achieves high Sharpe ratios during secular bull markets purely through market exposure.

Calibration The degree to which a model’s predicted probabilities match empirical outcome frequencies. A well-calibrated model that assigns 70% probability to an event will see that event occur approximately 70% of the time. Distinct from *refinement* (discrimination).

Calibration–discrimination trade-off The structural consequence of retraining a classifier on new data without a simultaneous recalibration step. Retraining shifts the decision boundary to better separate classes (improving refinement) while leaving the probability scale anchored to the pre-shift distribution (degrading calibration). Log loss penalises the resulting overconfidence even as directional accuracy improves.

Calmar ratio Annualised mean return divided by maximum drawdown. Captures whether returns were earned without catastrophic interim losses. A higher value indicates better return per unit of peak-to-trough risk.

Coefficient of Variation (CV) Standard deviation divided by the mean. Measures relative variability; used in the sensitivity analysis to assess how stable the accuracy delta is across the hyperparameter grid. CV below 3 is treated as indicating robustness.

Cohen’s h A measure of the practical difference between two proportions, analogous to Cohen’s d for means. $h \approx 0.07$ is considered a small effect; $h \geq 0.2$ is the conventional small-effect threshold.

Composite drift index ($\mathcal{D}_t$) A weighted aggregate of three normalised drift scores: feature distribution score (s_{feature} , weight 0.4), prediction drift score ($s_{\text{prediction}}$, weight 0.3), and performance drift score ($s_{\text{performance}}$, weight 0.3). Smoothed by EMA with $\alpha = 0.10$ before threshold comparison.

Concept drift (strict) A change in the conditional distribution $P(y \mid \mathbf{x})$: the predictive relationship between features and the target changes even if the feature distribution itself is stable. The most damaging drift type because it directly invalidates the learnt mapping.

Conviction gate See *position gate*.

- Cooldown period** A suppression window of 5 trading days after any retrain event during which further alarms are ignored. Prevents redundant retrains during post-drift turbulence. Also satisfies the T+2 settlement cycle buffer under MiFID II.
- Covariate shift** A change in the marginal distribution of input features $P(\mathbf{x})$ without a corresponding change in the conditional distribution $P(y | \mathbf{x})$. The most common early-warning signal in financial markets, preceding accuracy degradation by a mean of 11 days. Invisible to error-based detectors such as ADWIN, DDM, and EDDM.
- CUSUM (Cumulative Sum)** A sequential change-detection technique that accumulates small deviations from a reference mean over time. A sustained shift eventually causes the cumulative sum to cross a threshold; a single spike does not. Forms the mathematical basis of the Page–Hinkley detector.
- CVaR (Conditional Value at Risk)** The mean loss on the worst $q\%$ of trading days, where $q=5\%$ here. A tail-risk measure that captures how bad the worst days actually were, independently of the distribution’s centre.
- DDD (Diversity for Dealing with Drift)** A dynamic ensemble method that maintains diversity between sub-ensembles trained on different data windows. Excluded for the same compatibility reasons as ARF.
- DDM (Drift Detection Method)** Gama et al.’s (2004) error-based detector that raises an alarm when the rolling error rate rises above a statistical threshold. Requires labels at detection time and is blind to covariate shift.
- Decision boundary** The hyperplane (or surface) in feature space at which a classifier’s predicted class switches from 0 to 1. Retraining shifts this boundary to better fit the new distribution; the probability scale may not shift correspondingly, causing calibration degradation.
- Detector ablation** A systematic experiment that evaluates all $2^4 = 16$ subsets of the four detectors (KS, PSI, JS, PH) to identify which configuration is Pareto optimal on accuracy and runtime.
- Discordant pairs** Pairs of observations (here, trading days) where one model is correct and the other wrong. McNemar’s test counts discordant pairs to assess whether one classifier is systematically superior.
- Distributional shift** Any change in the joint distribution $P(\mathbf{x}, y)$ over time. Encompasses covariate shift, concept drift, and prior probability shift. The general term for what the detector stack is designed to identify.
- DWM (Dynamic Weighted Majority)** A dynamic ensemble method that weights component classifiers by recent accuracy. Excluded because it adapts through ensemble diversity rather than the discrete retrain triggers whose contribution this project is designed to measure.

EDDM (Early Drift Detection Method) Baena-García et al.’s (2006) extension of DDM.

Detects gradual drift by monitoring the rate of change of the error rate rather than its level. Shares DDM’s label dependency and covariate-shift blindness.

Efficient Markets Hypothesis (EMH) Fama’s (1970) hypothesis that all available information is already reflected in asset prices, leaving no exploitable patterns. The Adaptive Markets Hypothesis (AMH) provides a more empirically consistent account for non-stationary markets.

EMA (Exponential Moving Average) A weighted average that applies exponentially declining weights to older observations. Controlled by smoothing parameter α . At $\alpha = 0.10$, the half-life is approximately 6.6 trading days. Applied to the composite drift index to suppress single-day artefacts.

False Positive Rate (FPR) The fraction of non-drift periods during which a detector raises a false alarm. Computed exactly on synthetic benchmarks where the true change-point position is known.

Feature distribution detector A detector that monitors the statistical properties of input features $\mathbf{x}$ without requiring labels. Enables drift detection before accuracy has degraded. KS, PSI, and JS are the three feature distribution detectors used here.

GARCH See *ARCH* / *GARCH*.

Golden cross A technical analysis signal in which a short-term moving average (here, 20-day) rises above a long-term moving average (here, 60-day). Used as a feature capturing upward trend alignment.

Gradient Boosted Machine (GBM) An ensemble classifier that reduces bias by iteratively fitting new trees to the residual errors of the current ensemble. Each new tree corrects the errors of the previous one. One of the three base classifiers used in the framework.

Herbster–Warmuth regret bound The result that a blended learner achieves expected regret $\mathcal{O}(\sqrt{T \ln N})$ over T time steps when blending N experts. Motivates retaining prior model parameters ($\gamma = 0.15$) rather than fully replacing them at each retrain event.

Hidden Markov Model (HMM) A statistical model in which an observable sequence is generated by an unobserved (latent) Markov chain. Hamilton (1989) applied HMMs to model the business cycle as a two-state process; the latent states correspond to expansion and contraction regimes.

Hybrid switching strategy A meta-strategy that selects between the adaptive and static models at each time step based on a rolling log-loss signal, rather than always using the adaptive model. Reduces log-loss degradation by 71% but does not eliminate it.

IC (Information Coefficient) The Spearman rank correlation between a model’s predicted probabilities and the realised binary outcomes. Measures forecasting skill across the full probability range, independently of the 0.5 decision threshold and of position sizing.

- ICIR (Information Coefficient Information Ratio)** The IC divided by its standard deviation across evaluation windows. Analogous to a Sharpe ratio for forecasting skill; higher values indicate more consistent skill.
- IQR (Interquartile Range)** The difference between the 75th and 25th percentiles of a distribution. Robust to outliers. Used by `RobustScaler` as the scaling denominator and by the position-gate formula to describe the spread of lead-time events.
- Isotonic regression** A non-parametric probability calibration method that fits a monotone piecewise-constant function to map raw model scores to calibrated probabilities. An alternative to Platt scaling for recalibrating classifier outputs after retraining.
- Jensen-Shannon (JS) divergence** A symmetric, bounded ($[0, \ln 2]$) measure of the divergence between two probability distributions. Applied here to the model’s output probability distributions to detect prediction drift. Preferred over KL divergence because symmetry and boundedness enable normalisation to $[0, 1]$.
- KL divergence (Kullback–Leibler divergence)** An asymmetric measure of how one probability distribution differs from a reference distribution. Not used directly because its asymmetry and unboundedness complicate aggregation; JS divergence is preferred.
- Kolmogorov–Smirnov (KS) test** A non-parametric two-sample test that measures the supremum of the absolute difference between two empirical cumulative distribution functions. Bounded in $[0, 1]$; sensitive to shifts in the tails. $\mathcal{O}(n \log n)$ computational complexity.
- Kruskal–Wallis test** A non-parametric one-way analysis of variance that tests whether the medians of three or more groups differ significantly. Used here to assess whether the accuracy delta differs across low, medium, and high volatility regimes ($H=84.35$, $p<0.0001$).
- Lead-time hypothesis** The hypothesis that upstream feature-distribution detectors (KS, PSI) fire earlier than the performance-based Page–Hinkley detector, providing a warning before accuracy has degraded. *Rejected* by the empirical lead-time analysis: PH fired first in 74.1% of matched alarm events, with a median lead of 0 days.
- Leptokurtic** Having heavier tails than a Gaussian distribution (excess kurtosis > 0). Financial daily returns are leptokurtic: extreme moves occur far more often than a normal distribution would predict. Motivates the use of BCa bootstrapping and `RobustScaler`.
- Log loss (cross-entropy loss)** A proper scoring rule that penalises confident wrong predictions more severely than uncertain wrong ones. Defined as $-[y \log \hat{p} + (1 - y) \log(1 - \hat{p})]$. Used as the loss signal monitored by the Page–Hinkley detector; also a primary evaluation metric.
- Logistic Regression (LR)** A linear classifier that models $P(y = 1 \mid \mathbf{x})$ via the logistic function. Included as an interpretable linear baseline against which the non-linear gains of tree-based models are measured.

Look-ahead bias The error of using information that would not have been available at the time of a prediction. The central pitfall in financial backtesting. Prevented here by four protocol rules: features use only close-of-day t data; train/test splits are strictly sequential; static parameters are frozen after training; adaptive retrains use only data available at trigger time.

Maximum drawdown The largest peak-to-trough percentage loss sustained over the evaluation period. A drawdown of -28.3% means the portfolio fell 28.3% from its highest value to its subsequent lowest value before recovering.

McNemar’s test A paired statistical test that counts discordant pairs between two classifiers: days where one model is correct and the other wrong. A significant result indicates one classifier is systematically better, without requiring the two models’ outputs to be independent.

MiFID II The Markets in Financial Instruments Directive (revised 2014). Article 5 mandates T+2 settlement for EU equity markets, providing a regulatory motivation for the 5-day cooldown period.

Minimum retrain rows (`min_retrain_rows`) The floor on the number of observations used in any single retrain event, set to 400 rows (≈ 19 months). An architectural constraint rather than a tunable hyperparameter: reducing it cannot fix the 2022 degradation because the problem is that pre-shift observations are forced into every retrain regardless of trigger timing.

Momentum Cumulative log return over a lookback window ($\tau \in \{5, 10, 20, 60\}$ days). A positive momentum value indicates a rising price trend. One of the most persistently informative features in equity return prediction.

OHLCV Open, High, Low, Close, Volume: the five fields comprising a standard daily market data record. The feature matrix is constructed exclusively from OHLCV data to avoid using information unavailable at the close of each trading day.

OOS rollback (out-of-sample rollback) After each retrain, the new model is validated on the 20 most recent rows held out from training. If accuracy falls more than 8 percentage points below the pre-retrain baseline, the controller reverts to the previous model snapshot.

Page–Hinkley (PH) detector A sequential change-point detector based on the CUSUM formulation. Accumulates deviations of the rolling log loss from a reference mean; raises an alarm when the cumulative sum exceeds a threshold. Operates in $\mathcal{O}(1)$ time and space per observation. Does not require labels. The minimum sufficient detector: PH alone is Pareto optimal on accuracy and runtime.

Pareto optimal A configuration is Pareto optimal if no alternative achieves a strictly better outcome on one objective without being strictly worse on at least one other. Here, PH only is Pareto optimal on the accuracy–runtime frontier: every other configuration that improves accuracy also increases runtime, and vice versa.

- Platt scaling** A post-hoc calibration method that fits a logistic regression on the raw model scores to produce calibrated probabilities. Recommended as a per-retrain recalibration step to address the calibration–discrimination trade-off.
- Population Stability Index (PSI)** A distribution comparison statistic developed in credit risk monitoring. Computes a weighted sum of bin-proportion differences across decile bins. Established threshold conventions: below 0.10 is stable, 0.10–0.25 is moderate shift, above 0.25 is significant. $\mathcal{O}(n)$ complexity.
- Position gate (conviction gate)** A continuous blending mechanism that weights the adaptive model’s prediction by $w_{\text{adapt}} = \text{clip}(0.60 \cdot \tilde{\mathcal{D}}_t + 0.40 \cdot \tilde{s}_{\text{perf},t}, 0.25, 0.90)$, increasing the adaptive model’s contribution when drift is detected. The static model always contributes at least 10%; the adaptive model always contributes at least 25%.
- Prior probability shift** A change in the unconditional label distribution $P(y)$ without a corresponding change in $P(\mathbf{x} \mid y)$. In financial markets, this corresponds to a sustained shift in the proportion of up days versus down days (e.g., a persistent bear market).
- Random Forest (RF)** An ensemble classifier that trains many decision trees on random subsets of the data (bagging) and averages their predictions. Reduces variance relative to a single tree. One of the three base classifiers used in the framework.
- Regime (market regime)** A qualitatively distinct state of the financial market characterised by a stable data-generating process. Hamilton (1989) estimates mean US recession durations of 4–6 months and expansion durations of 10–12 months. A model trained within one regime may perform poorly when the regime changes.
- Regret bound** A theoretical guarantee on the cumulative difference between a learner’s performance and the performance of the best fixed strategy in hindsight. The Herbster–Warmuth bound of $\mathcal{O}(\sqrt{T \ln N})$ motivates the blended (rather than fully replaced) update strategy.
- RobustScaler** A feature scaling transformer that centres each feature on its median and scales by its interquartile range. Resistant to the extreme outliers present in financial return data.
- Rolling window** A fixed-length buffer of the most recent w observations used as the current reference distribution for detector comparisons. Set to $w=100$ days here.
- SGD (Stochastic Gradient Descent)** An online optimisation algorithm that updates model parameters using the gradient computed on a single observation (or small batch) rather than the full dataset. Used in the SGD weighted update strategy, where it maintains an online logistic regression component updated at each time step.
- Sharpe ratio** Annualised mean return divided by annualised return standard deviation, with the risk-free rate set to zero. Measures return earned per unit of total volatility. ΔSharpe is the adaptive model’s Sharpe minus the static model’s Sharpe.

SMA crossover (Simple Moving Average crossover) A financial trading rule that generates a buy signal when a short-term moving average crosses above a long-term moving average (golden cross) and a sell signal for the reverse. Used as a financial baseline (20-day and 50-day windows).

Sortino ratio Like the Sharpe ratio, but uses only the standard deviation of *negative* returns in the denominator. Penalises downside volatility without penalising upside volatility.

Spearman rank correlation A non-parametric correlation coefficient computed on the ranks of two variables rather than their raw values. Used in the Information Coefficient to measure the association between predicted probability ranks and realised outcome ranks, independently of the probability scale.

Stability–plasticity trade-off The fundamental tension in continual learning between retaining knowledge from the past (stability) and incorporating information about the present (plasticity). The blend weight $\gamma = 0.15$ positions the SGD update strategy toward the stable end: 85% of prior parameters are retained at each update.

Stationarity The property that the joint distribution $P(\mathbf{x}_t, y_{t+1})$ does not change over time. Assumed by most standard supervised learning frameworks; empirically untenable in financial markets.

Survivorship bias The distortion introduced when an evaluation corpus includes only instruments that survived to the end of the study period. Surviving instruments are on average more successful than the full historical population. Acknowledged as a limitation; a point-in-time constituent dataset would be required for a fully bias-corrected evaluation.

Tiered controller The framework component that maps the smoothed composite drift index to one of four actions: no action (below 60th percentile), SGD update (60th–75th percentile), partial retrain (75th–90th percentile), or full retrain (above 90th percentile). Severity thresholds are asset-specific, computed from each instrument’s reference window.

TPR (True Positive Rate) / Recall The fraction of true drift events detected within the alarm window. Computed exactly on synthetic benchmarks where the change-point position is known. Synonymous with recall in binary classification.

Transfer learning A machine learning technique in which a model trained on one task or dataset is adapted to a related task or dataset, rather than training from scratch. Identified as a future-work direction for data-sparse instruments such as international equities.

Volatility clustering The empirical regularity that large price moves (of either sign) tend to be followed by more large moves, and small moves by more small moves. Formally captured by ARCH/GARCH models.

Volatility targeting A position-sizing technique that scales the position weight inversely with realised volatility, keeping the portfolio’s risk contribution approximately constant over time. Identified as a future-work direction to replace the current drift-proportional conviction gate and cap maximum drawdown.

Volume Z-score The standardised deviation of today's trading volume from its rolling mean, measuring how unusual current trading activity is relative to recent history. High volume during a price move tends to signal institutional participation and a sustained regime transition.